BIO-SBI

Simulation experiments for planning organoid lineage experiments

Two-type branching models (LGR5$^+$ = A, KRT20$^+$ = B), neural posterior estimation, identifiability and robustness

Status of results	6 November 2026
Repository	`BIO-SBI`, branch `main`, commit `12c0827`
Data	synthetic only (no measured data used)
Audience	group planning the wet-lab organoid experiments

How to read this report. The executive summary (p. 1) is self-contained. Section 1 defines the model and every diagnostic. Each experiment section has the same four parts: *Question*, *Setup*, *Result*, *Comment*. Orange boxes mark experiments that were planned but never run. Parameters: A→AA etc. are outcome probabilities per cell event; τ_A, τ_B are mean cell lifetimes (Bellman–Harris only).

Executive summary

Scope. All simulation sweeps of E0–E4 are finished (966 runs), all on synthetic data from one reference parameter vector $\theta^\star$ (Table 1), 5 seeds per setting. Galton–Watson (GW, time in generations; mapping to days nominal) and Bellman–Harris (BH, time in hours; real-like design = days 6 and 10 = 144/240 h with 165/101 organoids). Every error is compared with the well-specified control of the same sweep.

Key findings

1. **The well-specified GW model is recovered**: at the real-like design MAE 0.033 over the 9 outcome probabilities, contraction 0.89, A→AA 0.52 (truth 0.55); across all E4 designs A→AA 0.47–0.54 — provided B's divisions are fixed to 0 (E0).
2. **B's row is prior-dominated if all transitions are free** (E0): B→A = 0.20 vs truth 0.80 at 5k–50k simulations, every design (uniform Dirichlet puts 0.03% mass above 0.8; a flat prior gives 0.53).
3. **Endpoints matter more than organoids, in both simulators** (E4): days/generations {4,6,10} beat every other set at every size; one late endpoint plateaus; a late extra endpoint ({6,10,14}) adds little. GW: contraction 0.80 → 0.95, MAE 0.052 → 0.023 (16 → 512 organoids/endpoint), every probability recovered by ≈128 (A→AB: 512). BH: 0.50 → 0.79, MAE 0.114 → 0.060.
4. **BH baseline bias = lifetime/self-renewal ridge.** Well-specified BH gives A→AA ≈ 0.32 and $\tau_A \approx$ 16 h (truth 0.55, 20 h), 0/5 covered, at every design. Fixing lifetimes to the truth gives 0.51 (5/5), MAE 0.109 → 0.059 (§6). More organoids do not help.
5. **A free time scale hides errors** (Figs. 1, 2). In BH the PPC stays ≤ 0.033 for every error except both deaths fixed wrong (0.096) and a global lifetime longer than A's (0.35–0.51). GW, without lifetimes, flags more: A death fixed too high (support ≈ 0), B death fixed at 4× truth (support 0.13–0.33), 40% multi-cell founders (PPC 0.12, support 0.16). Still silent in GW: wrong founder type, B death at 2× truth (B→A 0/5 covered), ignored mis-typing up to 2 posterior sd.
6. **Size of the tested errors (vs paired control; BH | GW):**
   - founders 50% B: A→AB −0.11, τ_B +4 h | shifts ≤ 0.07 (silent in both);
   - 40% multi-cell clones: τ_A −1.8 h, B→A −0.07 | A→AA +0.13, B→A −0.25 (0/5 covered, flagged);
   - B death fixed at 0.4 (true 0.1): B→A −0.19 | −0.23 to −0.26; A death fixed at 0.2 (true 0.03): A persist −0.10, τ_A −2 to −3 h | no fit; deaths fixed at 0: ≤ 0.07 in both (silent);
   - mis-typing ignored, $e = 0.2$: A→AB +0.09 (0.7 sd) | +0.15 (2.1 sd; already 1.4 sd at $e = 0.1$); modelled: information loss (A→AB contraction 0.30 → −0.22 | 0.85 → 0.37);
   - BH only: lifetime CV 0.18–0.5 harmless, exponential A→AA +0.16, B→A −0.18; quiescence (A persist) not identifiable.
7. **Founder composition**: founder type not learnable in BH (contraction ≤ 0.12), partly in GW (0.68 for truth 0.5); inferring it removes no shift. Founder size: stays at prior in BH; in GW it tracks the truth (0.71 for $p_1 = 0.6$) and inferring it removes the misfit and halves the shifts.
8. **Observed-organoid fraction** (detected/seeded): 0.94–0.95 at $\theta^\star$; dominated by A death (0.3 ⇒ 0.54–0.58); GW exact and BH simulated agree to 0.02 on average; A and B death trade off (one constraint, not two).
9. **Hidden third type (GW, E2c): unresolved.** Ignoring C is flagged (PPC 0.17–0.41, support ≈ 0), but modelling C — even with the true data model — fails worse (support ≈ 0, PPC 0.26–0.84): the truth (A→AA 0.79–0.905) has prior probability 10^{-6}–10^{-10} under the three-type prior. The misfit is a prior-tail effect and cannot be attributed to C.

Recommendations for the wet-lab design

- **Endpoints:** three, including an early one — e.g. days 4, 6, 10 (best in both simulators). A single day-10 endpoint is the weakest design.
- **Organoids:** ≈128 per endpoint. In GW this recovers every outcome probability with {4,6,10} (A→BB from 32, B persist from 64, B→A from 128; A→AB needs 512). In BH it gives contraction 0.71; A→BB and B death need ≥ 128, A→AB ≥ 256; self-renewal is not recovered at any size without lifetime information.
- **Measure independently:**
  (a) *A (LGR5⁺) division time* by live imaging — the single most valuable measurement (it is what separates BH, MAE 0.109, from the GW target, 0.033); per type, not pooled;
  (b) *seeding*: number of organoids seeded per well (observed fraction, sensitive to A death) and whether wells contain multi-cell seeds (≳15% multi-cell clones bias self-renewal and B→A in GW; then fix or infer the founder-size distribution);
  (c) *mis-typing rate* (2D vs 3D comparison) and pass it to the model: ignored, it is tolerable only up to $e \approx 0.05$ once posteriors are as narrow as in GW;

(d) *cell-cycle variability*: only whether CV is < 0.5 or near-exponential;
(e) *founder type* and *quiescence* (e.g. Ki67) if they matter biologically — the counts cannot provide them;
(f) *apoptosis* only if death rates are to be fixed; otherwise keep them free (a value too high biases B→A; in BH silently).

- **Biological constraints help more than data**: B divisions = 0 (if KRT20$^+$ cells do not divide) is what makes B→A identifiable.
- **Hidden third type**: no recommendation yet; E2c must be repeated with a truth inside the prior bulk (or an informative prior) first.

Methodological lessons

- **Prior tails**: a truth with $< 2\%$ prior mass (B→A = 0.8, A→AA = 0.55 under uniform Dirichlet) is pulled to the prior; with 10^{-6} or less (E2c three-type) amortised NPE extrapolates outside the prior support. SBC still looks fine. Report prior quantiles of the truth; check prior sensitivity.
- **Ridges and free time scales**: endpoint counts fix the growth rate; lifetime, self-renewal and quiescence trade off along it (posterior corr. +0.7, −0.6). The prior picks the point, and the free lifetime also absorbs (hides) errors that GW detects (E1a number, E1c).
- **Compare with paired controls**, not with the truth (BH baseline is biased); 5 seeds resolve shifts of ≈ 0.02–0.03 (probabilities) and 0.5–1.5 h.
- **Diagnostics**: SBC checks the inference model only; PPC and support mass are relative checks and vary between well-specified GW runs (support 0.07–0.89, PPC up to 0.10); support mass ≈ 0 = the model cannot explain the data (numbers from such runs are not estimates).

Planned, never run (Part II): X2 seed spread; X1 methods comparison, X3 GW fit to BH data, X4 combined errors, R real CRC0322 data, P0 BH timing; also E2c with a truth inside the prior bulk and asymmetric mistyping.

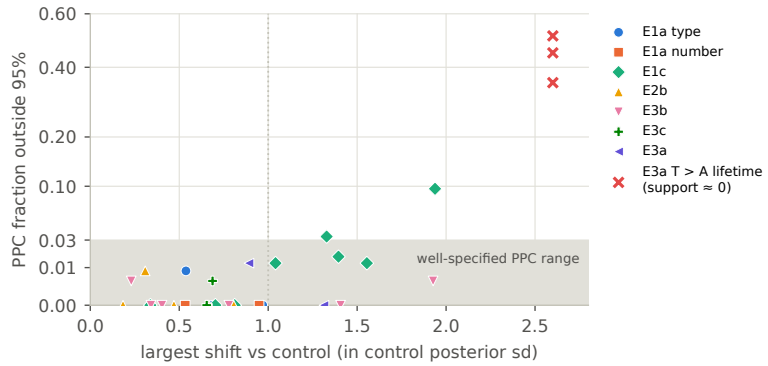


Figure 1: Detectability of BH misspecifications. Each point is one misspecified setting (mean of 5 seeds): largest shift of a posterior mean from its paired well-specified control, in units of the control's posterior sd, against the PPC. Grey band = PPC range of well-specified runs. Crosses: global lifetime $T > \tau_A$ (support mass ≈ 0).

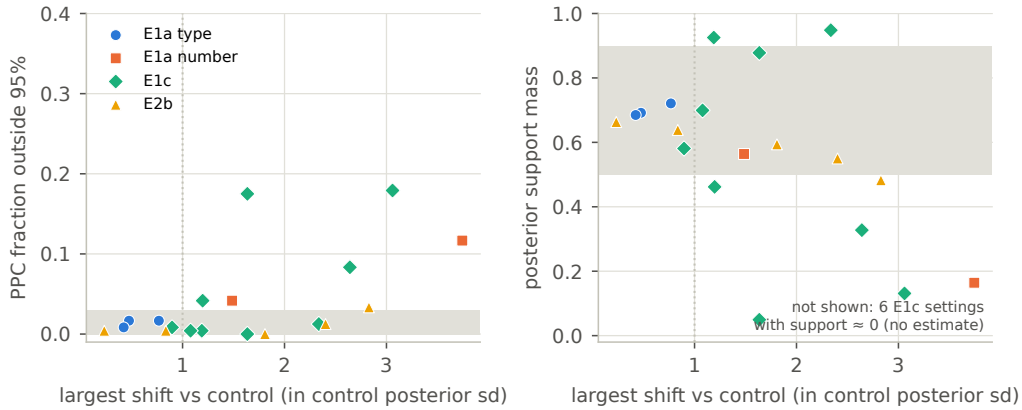


Figure 2: Detectability of GW misspecifications (E1a type and number, E1c, E2b): largest shift from the paired control in control posterior sd, against the PPC (left) and the posterior support mass (right). Grey bands = well-specified ranges. GW flags more errors than BH (no free time scale), but several shifts of 1–3 sd remain inside the well-specified bands.

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1 Methods overview

1.1 Model

- **Two cell types**: A = LGR5$^+$ (stem-like), B = KRT20$^+$ (differentiated).
- **One event per cell**: a cell is replaced by 0, 1 or 2 daughters. Outcomes of A: death, persist (stays, no division), A→B (switch), divisions A→AA, A→AB, A→BB. B mirrors A.
- **Parameters**: one probability simplex per type; Dirichlet(1,...,1) prior (uniform).
- **Laws**: `two_type` = all 12 outcomes free (10 free params); `reversible_ab` = B divisions fixed to 0 (B: death, persist, B→A; 7 free) — the baseline of every experiment after E0.
- **Founders**: a clone (organoid) starts from N cells ($P(N{=}n)$ = `founder_size`) of i.i.d. type (`founder`); baseline: one A cell.

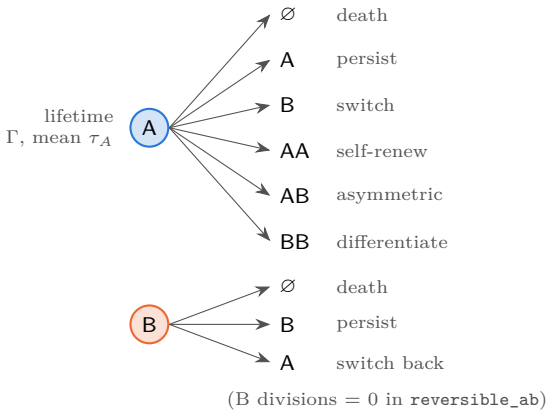


Table 1: Reference truth $\theta^\star$ used by every experiment unless stated (`law.reference`); one founder cell of type A. Prior probability of exceeding the truth under the uniform Dirichlet: A→AA 1.8%; B→A 0.03% (`two_type`) or 4% (`reversible_ab`).

Type	death	persist	switch	→AA	→AB	→BB	mean lifetime (BH)
A	0.03	0.03	0.05 (A→B)	**0.55**	0.05	0.29	τ_A = 20 h
B	0.10	0.10	**0.80** (B→A)	0	0	0	τ_B = 30 h

BH lifetimes: Gamma, shape 8 (CV 0.35). GW: one event per generation.

1.2 Two simulators

	Galton–Watson (GW)	**Bellman–Harris (BH)**
Time	synchronous generations; endpoints in *generations* (real-like: 6 and 10)	continuous; Gamma lifetimes per type; endpoints in *hours* (real-like: 144 and 240 h = days 6, 10)
Extra parameters	none	τ_A, τ_B (log-uniform prior); shape fixed (8)
Sampling	exact detection-conditioned sampler; exact observed fraction	forward simulation + rejection; observed fraction estimated
Cost	cheap	slower (rejection sampling; P0 timing not run)

GW and BH share the outcome law but not the time unit; a GW generation is *not* a day (X3 needs a mapping).

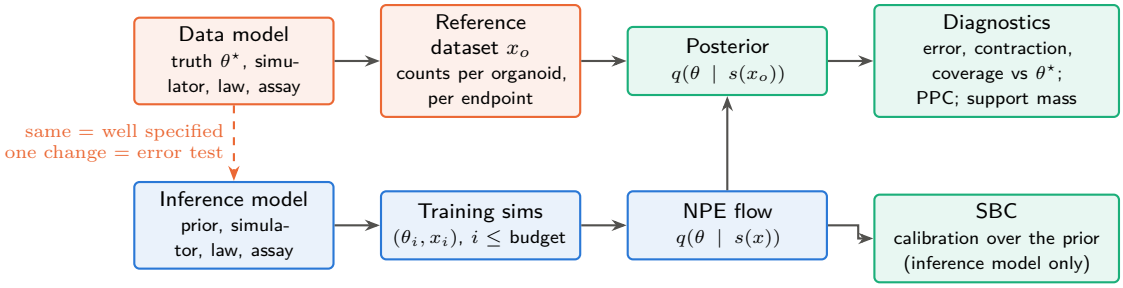
1.3 Observation, data model and inference model

- **Observation**: biology → capture (Binomial per true type) → labelling (confusion matrix true type × label; may be rectangular, e.g. hidden C read as A with prob. q) → selection of *recovered* clones (≥ 1 detected cell). Baseline: perfect capture and typing.
- **Data**: per endpoint, A and B counts of each recovered organoid. Endpoints are destructive: different organoids per endpoint.
- **Misspecification**: the reference data come from a separate *data model* (`data.simulator`, `data.law`, `data.observation`, with seed `data_seed`); the *inference model* defines prior, training simulations and checks. An error is tested by changing one side only.

1.4 Inference

- **NPE** (`sbi` 0.27, neural spline flow, 5 transforms, width 64), amortised, 1 round; sequential NPE (SNPE) only in E0.
- **Summaries** (`moments`): per endpoint, mean, sd, quantiles of $\log(1+n)$ for A, B, total; fraction of clones containing each type; fraction empty; correlation of log counts.
- **Budgets**: `dev` = 5 000 simulations (all BH and most GW error experiments); 20 000 for E0/design comparison; E0 also 50 000. E0 showed 5k→50k does not change recovery. Per run: 2 000 posterior samples, SBC on 200 prior draws, PPC with 200 datasets.
- **Parameterisation**: stick-breaking of each simplex (exact Dirichlet reparameterisation).

1.5 Pipeline and experiment logic



Experiment logic: vary one error on the data side (or one assumption on the inference side), keep everything else at the baseline, and compare with the *well-specified control of the same sweep* (same seeds, same datasets; often the same trained network, so differences are paired by seed).

1.6 Diagnostics and how to read them

Metric	Definition	How to read
Posterior mean / error	mean of 2 000 posterior samples; |mean − truth|. MAE over probabilities only (`mae_probabilities`); lifetimes separately	compare misspecified vs control; "shift" = difference of posterior means, paired by seed, ± paired SE over 5 seeds
Contraction	$1 - \mathrm{Var}_{\mathrm{post}}/\mathrm{Var}_{\mathrm{prior}}$	0 = data uninformative; < 0 = wider than prior; > 0.5 = informative
90% coverage across seeds	number of seeds (of 5) whose 90% credible interval contains the truth	4–5/5 expected; 0/5 with high contraction = confident bias
SBC coverage error	mean |empirical − nominal| coverage of central intervals (50–95%) over 200 prior draws	≤ 0.03 = calibrated. Averages over the prior: blind to tail truths and to misspecification
PPC fraction outside 95%	share of summary features whose observed value lies outside the 95% posterior predictive interval	well specified: 0–0.03. Higher = the model cannot reproduce the data. Relative check
Posterior support mass	share of the NPE posterior inside the prior support	well specified ≈ 0.5–0.9 (not 1). ≈ 0 = model cannot explain the data; few/no samples, not an estimate
Recovered (TODO criterion)	contraction > 0.5, |error| < 0.05, coverage ≥ 4/5	used for "organoids needed" (E4)

Reading rules used throughout. (1) *BH baseline bias*: well-specified BH runs give A→AA ≈ 0.32 and $\tau_A \approx$ 16 h (truth 0.55, 20 h); BH misspecified arms are compared with their control, never with the truth. (2) With all transitions free, B's row is prior-dominated (E0). (3) A misspecification that raises neither PPC nor lowers support mass is *undetectable from the fit*. (4) 5 seeds per setting: effects below ≈ 2 paired SE are not resolved. (5) Wall times in run metrics were measured under heavy contention; only the device benchmark (§20) measures speed.

Part I

Finished experiments

2 E0 — Identifiability of B's row (GW)

Question

With all transitions free, B→A (true 0.80) came out at ≈0.2. Is this under-training, one bad dataset, miscalibration, a prior-tail effect, or non-identifiability (switching vs division)?

Setup

Simulator	Galton–Watson, perfect observation
Law (free)	`two_type` (10) vs `reversible_ab` (7; B divisions fixed to their true 0)
Data vs inference	well specified (same $\theta^\star$, Table 1)
Design	generations 4, 8, 12 × 32 clones (96 organoids)
Inference / budget	NPE amortised 5k, 20k, 50k; SNPE 4 rounds × 5k; `two_type` with flat prior on B→A (Beta(1,1)), 20k
Seeds / runs	5 / 40 (+ direct checks: C2ST of mean-preserving alternatives, 500 datasets per side)
Varied	law, budget, method, prior
Folder	`results/b_row/` (`distinguish.csv`, `truth_vs_posterior_20k*.csv`)

Result

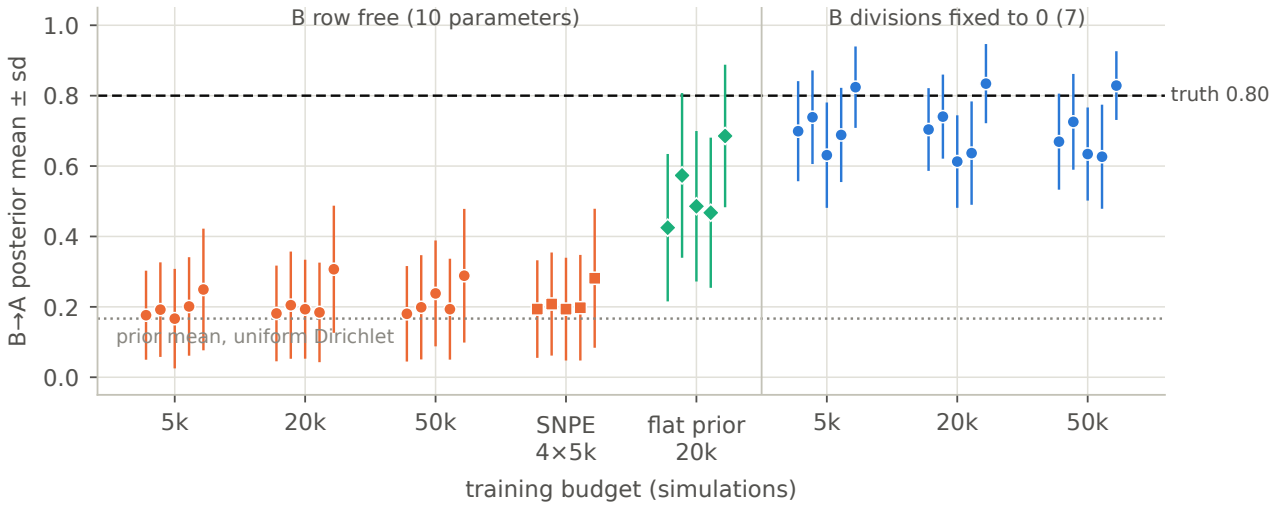


Figure 3: B→A posterior mean ± sd per seed. Orange: all B outcomes free (uniform Dirichlet; squares: SNPE). Green: flat prior on B→A. Blue: B divisions fixed to 0.

Law	Training	B→A mean	post. sd	contraction	covered	MAE (all)
`two_type`	5k / 20k / 50k	0.20 / 0.21 / 0.22	0.14–0.15	−0.03 / −0.13 / −0.18	0/15	0.14
`two_type`	SNPE 4×5k	0.21	0.16	−0.22	0/5	0.14
`two_type`, flat prior	20k	0.53	0.21	0.46	4/5	0.10
`reversible_ab`	5k / 20k / 50k	0.72 / 0.71 / 0.70	0.13–0.14	0.67 / 0.71 / 0.69	15/15	0.05

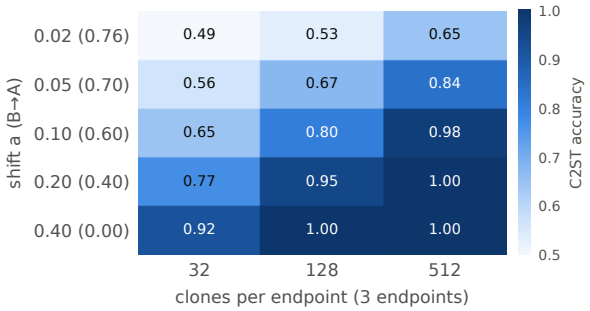


Figure 4: C2ST accuracy separating the truth from a mean-preserving alternative (B→AA = a, B→A = $0.8 - 2a$, B death = $0.1 + a$). 0.5 = indistinguishable.

- SBC coverage error 0.02–0.04 in every arm (looks calibrated).
- PPC ≤ 0.014 in every arm.
- Flat prior: corr(B→A, B→AA) = −0.67; SBC fails (0.069) for B's other coordinates (concentration 0.2), not for B→A.
- Truth-vs-posterior C2ST: inconclusive (0.62–0.95 in both laws).

Comment

- Under-training, dataset and calibration are ruled out: 10× more simulations, 5 datasets, SBC all unchanged.
- Main cause = prior tail: B→A ∼ Beta(1,5), P(> 0.8) = 0.03% (≈ 15 of 50k training simulations). SNPE cannot help: round 0 is already prior-like.
- Data weakly confound switching with division: even with a flat prior the posterior is wide (sd 0.21) and anti-correlated with B→AA. They differ only in clone-size variability; 128–512 clones/endpoint make moderate differences detectable.
- Consequence: fix what biology allows (B divisions = 0 ⇒ B→A ≈ 0.70, contraction 0.7); otherwise report prior sensitivity. All later experiments use `reversible_ab`.

3 Design comparison — one vs three endpoints (GW)

Question

Which planned design identifies the model best, and is it the number of time points or the number of organoids that matters?

Setup

Simulator	Galton–Watson, perfect observation
Law (free)	`two_type` (10) and `reversible_ab` (7)
Data vs inference	well specified, $\theta^\star$
Design	`single_128`: gen. 12 × 128; `three_128`: gen. 4, 8, 12 × 128; `single_384`: gen. 12 × 384; reference `three_32` (from E0)
Inference / budget	NPE amortised, 20k
Seeds / runs	5 / 30 (+ 10 reference runs from E0)
Varied	design × law
Folder	`results/design_comparison/`

Result

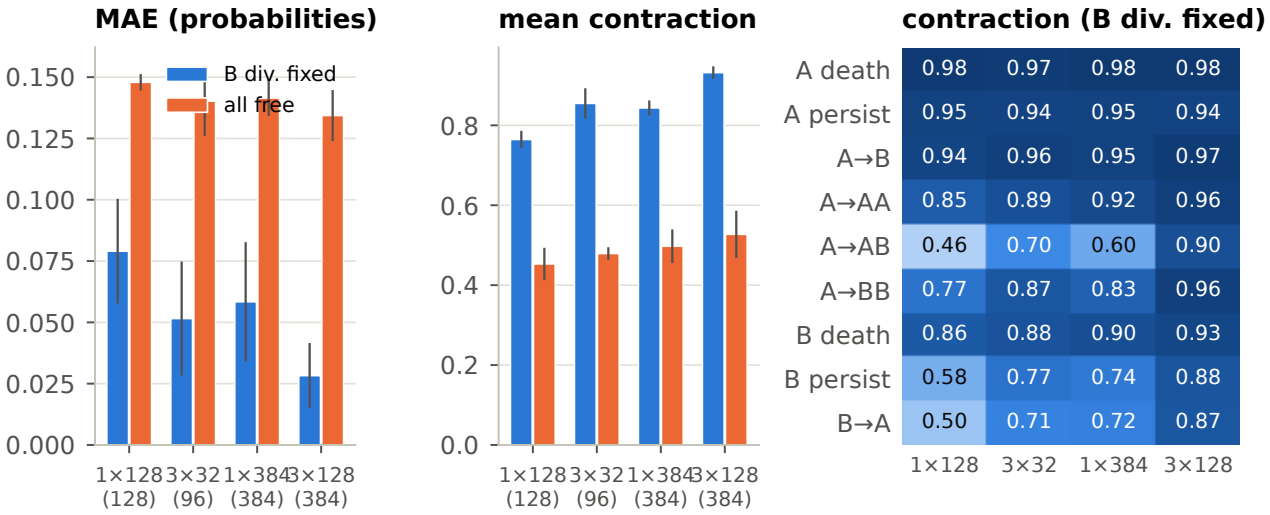


Figure 5: Left, centre: MAE and mean contraction per design (mean ± sd over 5 seeds); total organoids in brackets. Right: contraction per parameter, B divisions fixed.

Design (`reversible_ab`)	organoids	MAE	contraction	90% cov.	B→A	B→A contr.	PPC
1 × 128	128	0.079	0.77	0.87	0.61 ± 0.17	0.50	0.000
3 × 32	96	0.052	0.86	0.96	0.71 ± 0.13	0.71	0.008
1 × 384	384	0.058	0.84	0.93	0.67 ± 0.12	0.72	0.000
3 × 128	384	**0.028**	**0.93**	0.93	**0.78 ± 0.09**	**0.87**	0.031
`two_type`, any design	96–384	0.13–0.15	0.45–0.53	0.69–0.78	0.18–0.25	≤ −0.03	≤ 0.014

Comment

- At equal organoids, 3 endpoints halve the error of 1 (0.028 vs 0.058). Gains are in dynamics: A→AB contraction 0.90 vs 0.60, B→A 0.87 vs 0.72.
- 96 organoids over 3 endpoints ≈ 384 at one endpoint.
- With all transitions free no design recovers B's row (B→A contraction ≤ 0) — consistent with E0.
- Calibration fine everywhere (SBC 0.02–0.03). Caveat: GW, one truth, generations not days.

4 E4 — Amount of data: organoids and endpoints (BH)

Question

How do contraction and error depend on organoids per endpoint (16–512) and on the measured days? What is the minimum number to recover each parameter?

Setup

Simulator	Bellman–Harris, Gamma lifetimes shape 8
Law (free)	`reversible_ab`: 7 probabilities + τ_A, τ_B (9)
Data vs inference	well specified, $\theta^\star$
Design	clones per endpoint $\in \{16, 32, 64, 128, 256, 512\}$ × endpoints {240}, {144, 240}, {144, 240, 336}, {96, 144, 240} h = days {10}, {6,10}, {6,10,14}, {4,6,10}
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 120
Varied	clones × endpoint set
Folder	`results/e4/bh/`

Result

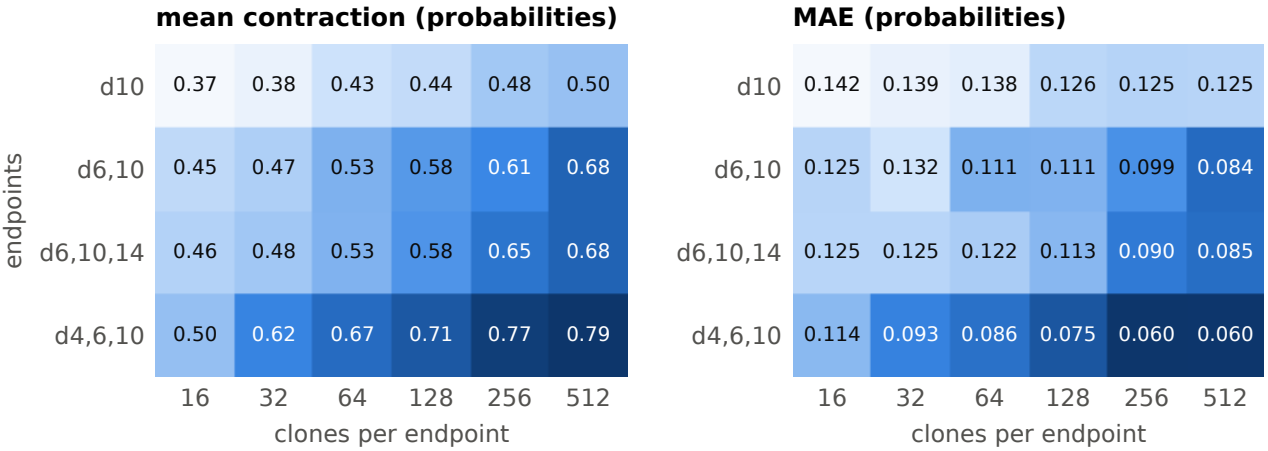


Figure 6: Mean contraction and MAE of the 7 outcome probabilities (mean over 5 seeds).

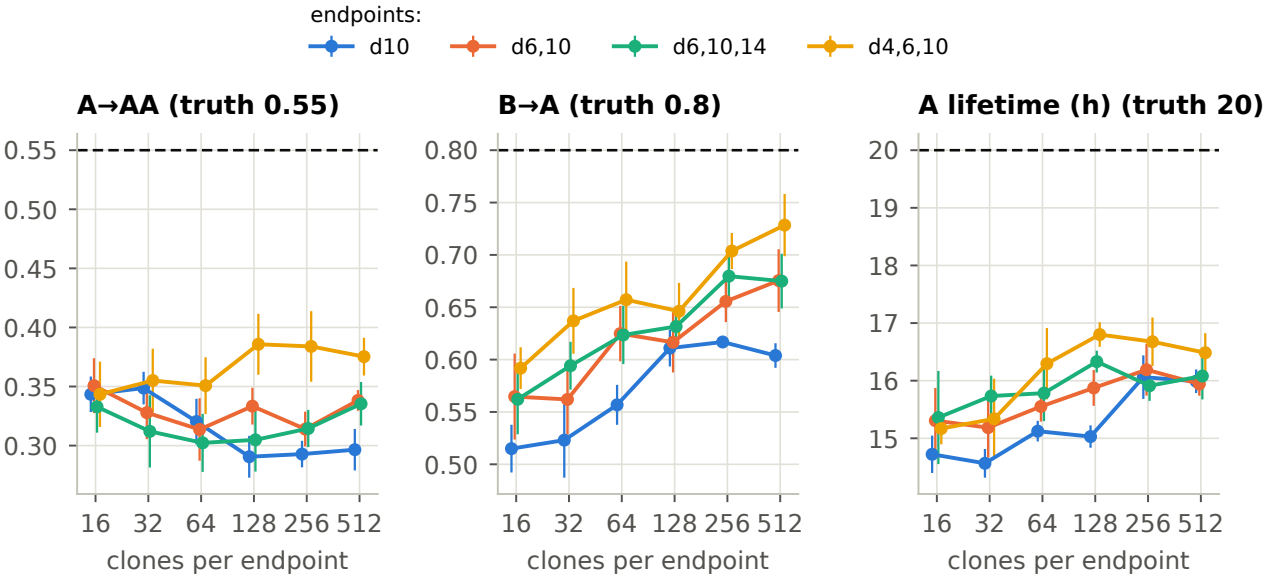


Figure 7: Posterior means (mean ± SE over seeds) vs organoids per endpoint, one line per endpoint set; dashed = truth.

Minimum organoids per endpoint for "recovered" (contraction > 0.5, $|\text{error}| < 0.05$, $\geq 4/5$ covered); — = not reached at 512.

Parameter	day 10	days 6, 10	days 6, 10, 14	days 4, 6, 10
A death, A→B (small truths)	16	16	16	16
A→BB	—	512	256	**128**
B death	—	256	256	**128**
A→AB	—	—	—	**256**
A→AA, A persist, B persist, B→A	—	—	—	—
τ_A, τ_B (biased: τ_A 15–17 h, 0/5 covered)	—	—	—	—

Comment

- An early endpoint (day 4) helps most; day 10 alone plateaus (contraction 0.37 → 0.50 from 16 to 512).
- With several endpoints more organoids keep helping (MAE 0.114 → 0.060 for days 4/6/10); B→A rises 0.59 → 0.73.
- A→AA (0.29–0.39) and τ_A (15–17 h) are never recovered; coverage falls to 0 as posteriors narrow on the wrong values: the ridge (§6), not lack of data.
- "Recovered" for A death / A→B is easy because the truths are small (|error| < 0.05 is loose there).
- Caveat: one truth, perfect assay, single A founders, 5k simulations; GW half and comparison in §5.

5 E4 — Amount of data (GW) and comparison with BH

Question

Same as §4 for Galton–Watson: minimum organoids per endpoint and best endpoint set when there is no lifetime ridge.

Setup

Simulator	Galton–Watson (endpoints in *generations*; day-mapping nominal)
Law (free)	`reversible_ab` (7 probabilities)
Data vs inference	well specified, $\theta^\star$
Design	clones per endpoint $\in \{16, \ldots, 512\}$ × generations {10}, {6,10}, {6,10,14}, {4,6,10}
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 120 (all with 2 000 posterior samples, 200 SBC cases)
Folder	`results/e4/gw/`

Result

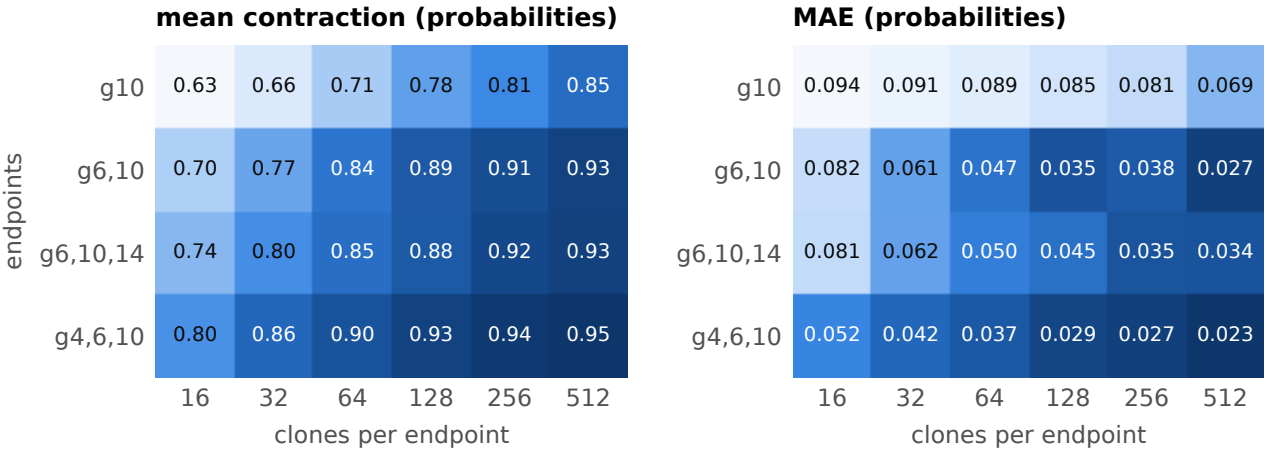


Figure 8: E4 GW: mean contraction and MAE of the outcome probabilities (mean over 5 seeds); endpoints in generations.

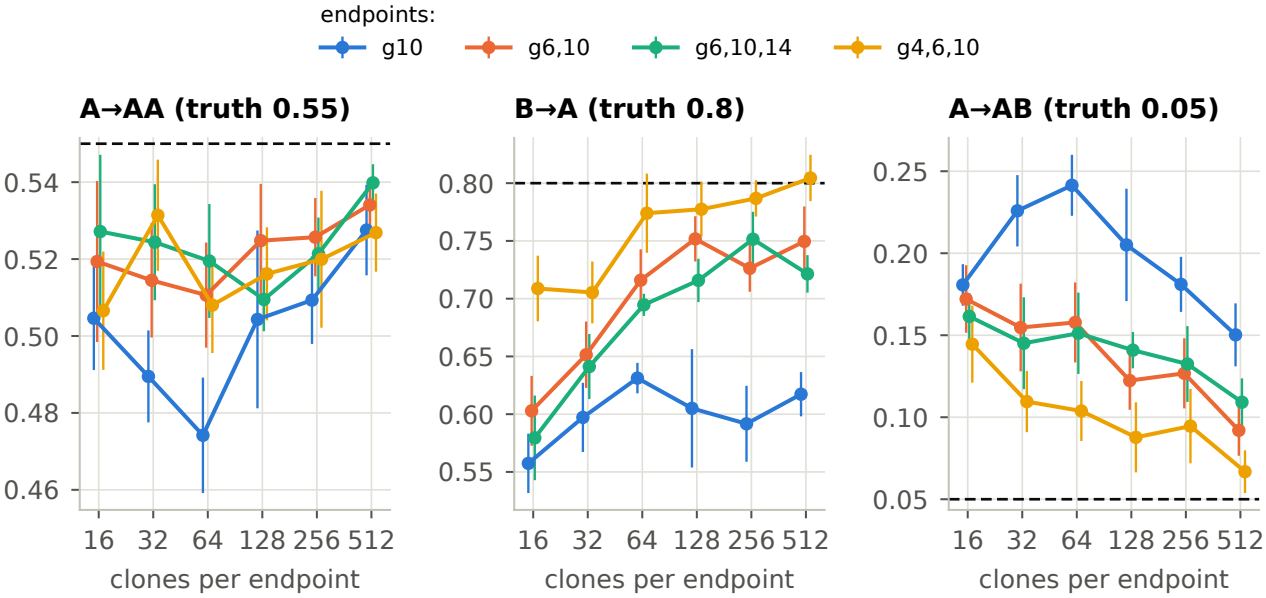


Figure 9: E4 GW: posterior means (mean ± SE over seeds) vs organoids per endpoint; dashed = truth.

Minimum organoids per endpoint for "recovered" (as in §4; must hold at every larger size; — = not reached)

	GW, generations				**BH**, days
Parameter	{10}	{6, 10}	{6, 10, 14}	{4, 6, 10}	{4, 6, 10}
A death, A persist, A→B	16	16	16	16	16 (A persist: —)
B death	32	16	16	16	128
A→AA	256	16	16	—[a]	—
A→BB	—	256	—	**32**	128
B persist	—	512	—	**64**	—
B→A	—	—	—	**128**	—
A→AB	—	512	—	512	256

[a] error ≤ 0.04 at every size, but coverage 2–4/5 (narrow posteriors, sd ≈ 0.02).

	BH (days)	**GW** (generations)
Contraction, 16 → 512, {4,6,10}	0.50 → 0.79	0.80 → 0.95
MAE, 16 → 512, {4,6,10}	0.114 → 0.060	0.052 → 0.023
Single late endpoint, contraction	0.37 → 0.50	0.63 → 0.85 (A→AB / A→BB confounded)
A→AA (0.55)	0.29–0.39, 0–2/5 covered	0.47–0.54 at every design
B→A (0.80), {4,6,10}, 128 / 512	0.65 / 0.73	0.78 / 0.80
Support mass / PPC (well specified, per run)	0.52–0.92 / ≤ 0.042	0.07–0.89 / ≤ 0.10 (group means ≤ 0.021)

Comment

- **Same design ranking as BH**: an early endpoint is worth most ({4,6,10} best at every size); a late extra endpoint ({6,10,14}) adds little over {6,10}; one endpoint plateaus and cannot separate A→AB from A→BB (0.15–0.24 vs 0.05; 0.15–0.20 vs 0.29) at any size. The ranking is a property of the endpoint information, not of the time model.
- **Without lifetimes everything is recoverable**: with {4,6,10} and ≈128 organoids per endpoint all probabilities except A→AB (512) meet the criterion; A→AA is right from 16 organoids. The GW numbers are the attainable target *if* division times are known (BH with lifetimes fixed: MAE 0.059, §6).
- Checks: single well-specified GW runs reach PPC 0.10 and support mass 0.07 ({4,6,10}, seed 4): borderline single values are not evidence of misspecification.
- Caveat: generations are not days; one truth, perfect assay, single A founders, 5k simulations. Planned extension (not run): `law=two_type` at 256–512 clones.

6 BH ridge — lifetime vs self-renewal

Question

Is the BH baseline bias (A→AA ≈ 0.32, $\tau_A \approx 16$ h) a ridge on which endpoint counts fix only a combination of lifetime and self-renewal, with the prior picking the point?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	control: `reversible_ab` (9); lifetimes fixed: 7; flat prior: 9
Data vs inference	data always $\theta^\star$; only the inference model changes
Design	real-like: 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Arms	control (from E3b, shape 8); τ_A=20, τ_B=30 h fixed; flat prior on A→AA (concentration 1 vs 0.2 for A's other outcomes: A→AA ∼ Beta(1,1))
Seeds / runs	5 / 10 (+5 control)
Folder	`results/bh_ridge/` (control: `results/e3b/*lifetime_shape=8*`)

Result

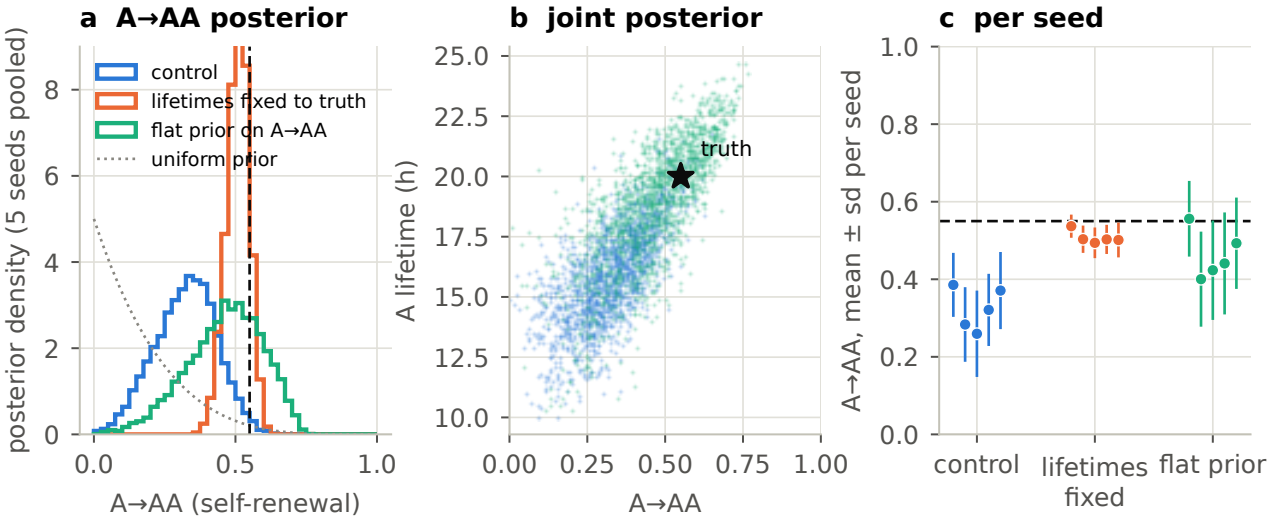


Figure 10: (a) A→AA posterior pooled over 5 seeds; dotted: uniform-Dirichlet prior Beta(1,5). (b) Joint posterior of A→AA and τ_A (blue: control; green: flat prior); star = truth. (c) Per-seed means ± sd; dashed = truth.

	Control	Lifetimes fixed	Flat prior
A→AA (0.55)	0.32 ± 0.10, 0/5, contr. 0.52	**0.51 ± 0.04, 5/5, 0.93**	0.46 ± 0.12, 5/5, 0.83
A→AB (0.05)	0.23, 3/5	0.17, 2/5	0.20, 5/5, contr. 0.00
B→A (0.80)	0.63, 5/5	0.68, 5/5	0.46, 3/5
τ_A (20 h)	15.9 ± 1.9, 0/5	fixed	18.9 ± 2.2, 5/5
MAE (probabilities)	0.109	**0.059**	0.117
SBC error / PPC / support	0.025 / 0.008 / n/r	0.031 / 0.025 / 0.65	0.079 / 0.000 / 0.51

Comment

- Ridge confirmed: posterior corr(A→AA, τ_A) ≈ +0.7 (E3c). Removing the lifetime degree of freedom recovers self-renewal and halves the error.
- The prior chose the point: only 1.8% of uniform-Dirichlet mass lies above 0.55. A flat prior re-centres (5/5 covered) but stays wide, degrades B→A and calibration: not a fix.
- GW has no lifetimes and no ridge: the GW control at the same design gives A→AA 0.52 (§7), and 0.47–0.54 at every E4 design (§5).
- Consequence: measure A's division time (live imaging) or use it as an informative prior; otherwise report BH results with prior sensitivity.

7 E1a — Wrong founder type (organoid death as initial conditions; BH and GW)

Question

If some organoids are founded by B cells (fraction $1 - f$) but the model assumes A founders, are the dynamics (A, B rows) biased? Does inferring the founder mix help?

Setup

Simulator	Bellman–Harris (shape 8) and Galton–Watson
Law (free)	`reversible_ab`: BH 9, GW 7; inferred arm adds `founder.A` (Dirichlet(1,1))
Data model	founder A with prob. $f \in \{1, 0.9, 0.7, 0.5\}$, else B; dynamics $\theta^\star$
Inference model	fixed arm: founder = A (wrong for $f < 1$); inferred arm: founder share free
Design	real-like: BH 144/240 h, GW generations 6/10; 165/101 organoids
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 80 (4 f × 2 arms × 2 simulators)
Control	$f = 1$ of the same arm (same trained network per seed: paired shifts)
Folder	`results/e1a/founder_type/{fixed,inferred}_{bh,gw}`

Result

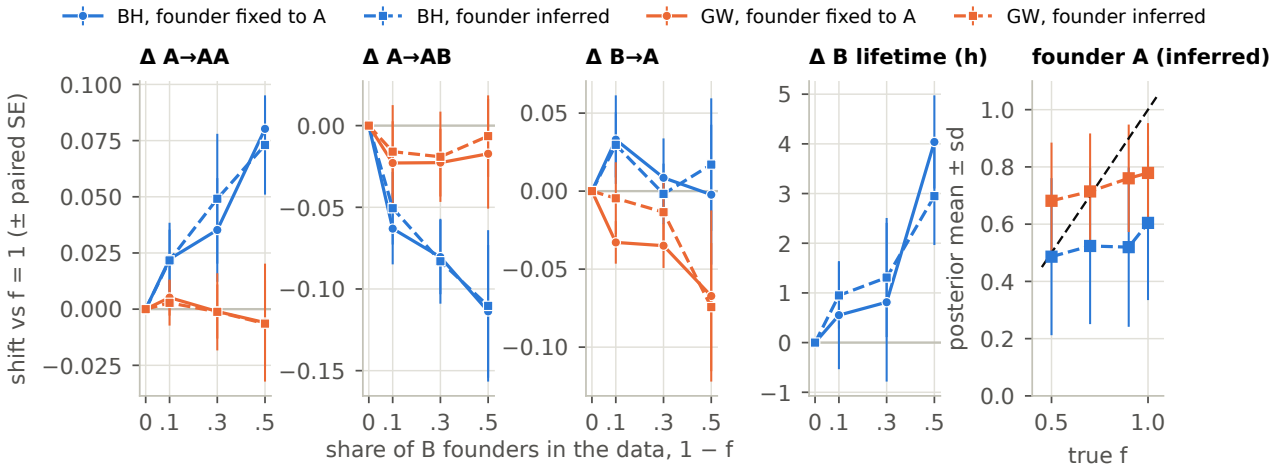


Figure 11: Shift of posterior means vs the $f = 1$ control (± paired SE over 5 seeds); blue BH, orange GW; solid = founder fixed to A, dashed = founder inferred. Right: inferred founder A share vs truth.

	BH, founder fixed				**GW**, founder fixed			
	control	$f = 0.9$	0.7	0.5	control	0.9	0.7	0.5
A→AA (0.55)	0.32 [0/5]	+0.02	+0.04	+0.08 ± .01	0.52 [3/5]	+0.01	−0.00	−0.01
A→AB (0.05)	0.23 [3/5]	−0.06	−0.08	−0.11± .04	0.12 [3/5]	−0.02	−0.02	−0.02
A→B (0.05)	0.07	+0.01	+0.01	+0.02 ± .006	0.035	+0.00	+0.01	+0.02 ± .006
B death (0.10)	0.17	−0.02	+0.00	−0.02	0.10	+0.01	+0.02	+0.03 ± .014
B→A (0.80)	0.63 [5/5]	+0.03	+0.01	−0.00	0.76 [5/5]	−0.03	−0.04	−0.07± .06
τ_B (30 h)	20.2	+0.6	+0.8	+4.0 ± 0.9			—	
MAE (probabilities)	.109	.090	.095	.089	.033	.033	.036	.047
PPC	.008	.008	0	0	.004	.008	.017	.017
support mass	n/r	n/r	n/r	n/r	0.67	0.69	0.69	0.72

Founder inferred: `founder.A` posterior mean (contraction) for $f = 1$ / 0.9 / 0.7 / 0.5:
BH 0.60 / 0.52 / 0.52 / 0.49 (0.06–0.12) GW 0.78 / 0.76 / 0.71 / 0.68 (0.49–0.62); GW support mass 0.39–0.59

Comment

- **Small shifts, PPC blind** in both simulators. BH: A→AB −0.11, A→AA +0.08 (towards the truth, by accident of the control's bias), τ_B +4 h at $f = 0.5$; all < 1 posterior sd. GW: shifts ≤ 0.07 (B→A, not resolved: ±0.06); A→B +0.02 and B death +0.03 resolved but ≈ 0.5 sd.
- **GW well-specified baseline is good**: MAE 0.033, contraction 0.89, A→AA 0.52 — the BH bias is specific to BH (ridge).
- **Founder mix**: not identifiable in BH (stays at prior 0.5); partly in GW (monotone but compressed: 0.68 at $f = 0.5$, 0.78 at $f = 1$). Inferring it removes none of the shifts in either simulator and costs τ_B (−3.9 h, BH).
- Plausible reason: B founders switch to A quickly (B→A = 0.8), so B- and A-founded clones look alike by day 6.
- Consequence: keep single-A founders unless measured at seeding; if measured, fix to the measured value. $f < 0.5$ not tested.

8 E1a — Wrong number of founder cells (BH)

Question

If some organoids start from 2–3 A cells (P(1 cell) = p_1) but the model assumes single-cell seeding, are the dynamics biased? Can the founder size be inferred?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	`reversible_ab` (9); inferred arm adds `founder_size` (sizes 1–3, Dirichlet(8,1,1))
Data model	founder size $\{1,2,3\}$ with $p_1 \in \{1, 0.85, 0.6\}$ ($\{.85,.12,.03\}$, $\{.6,.3,.1\}$), all A
Inference model	sizes [1] (wrong for $p_1 < 1$) or [1,2,3] inferred
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 30
Folder	`results/e1a/founder_number_bh/` (GW: §9)

Result

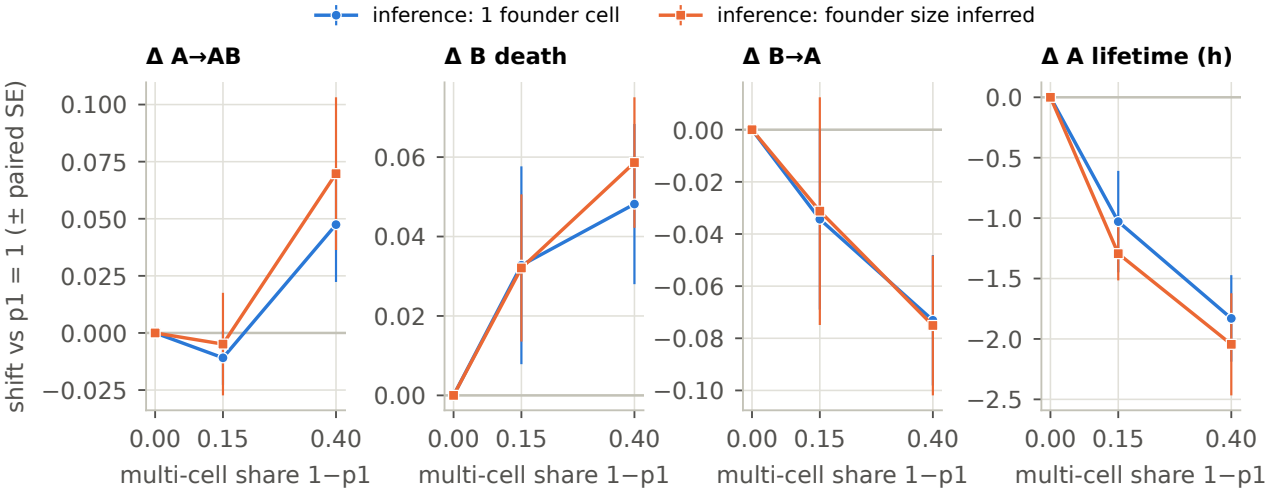


Figure 12: Shift vs the $p_1 = 1$ control of the same arm (± paired SE).

Arm, p_1	A→AB (.05)	B death (.10)	B→A (.80)	τ_A (20 h)	`founder_size.1`
[1], control	0.23	0.17	0.63 [5/5]	15.9	fixed
[1], 0.85 / 0.6	−0.01 / +0.05	+0.03 / +0.05 ± .02	−0.03 / −0.07 ± .03	−1.0 / −1.8 ± 0.4	fixed
[1,2,3], 0.85 / 0.6	−0.00 / +0.07	+0.03 / +0.06	−0.03 / −0.08	−1.3 / −2.0	0.81 / 0.78 (contr. ≤ 0.11)

Comment

- 40% multi-cell clones shift mostly lifetimes and B's row; coverage unchanged; PPC ≤ 0.008 (silent).
- Founder size stays at its prior (0.78–0.83); inferring it does not remove the shifts.
- Observed fraction rises with multi-cell seeding (0.95 → 0.97–0.98): fewer clones go extinct.
- Consequence: tolerable at $p_1 \geq 0.6$; do not infer founder size; measure it at seeding if needed. No support mass recorded (pre-`43e9e44` code).

9 E1a — Wrong number of founder cells (GW)

Question

As §8 for Galton–Watson: are the dynamics biased by multi-cell founders, can the founder size be inferred, and does the answer depend on the time model?

Setup

Simulator	Galton–Watson
Law (free)	`reversible_ab` (7); inferred arm adds `founder_size` (sizes 1–3, Dirichlet(8,1,1))
Data model	founder size $\{1,2,3\}$, $p_1 \in \{1, 0.85, 0.6\}$, all A
Design	real-like generations 6/10 × 165/101
Inference / budget	NPE amortised, 5k (one network per arm and seed: paired shifts)
Seeds / runs	5 / 30
Folder	`results/e1a/founder_number_gw/`

Result

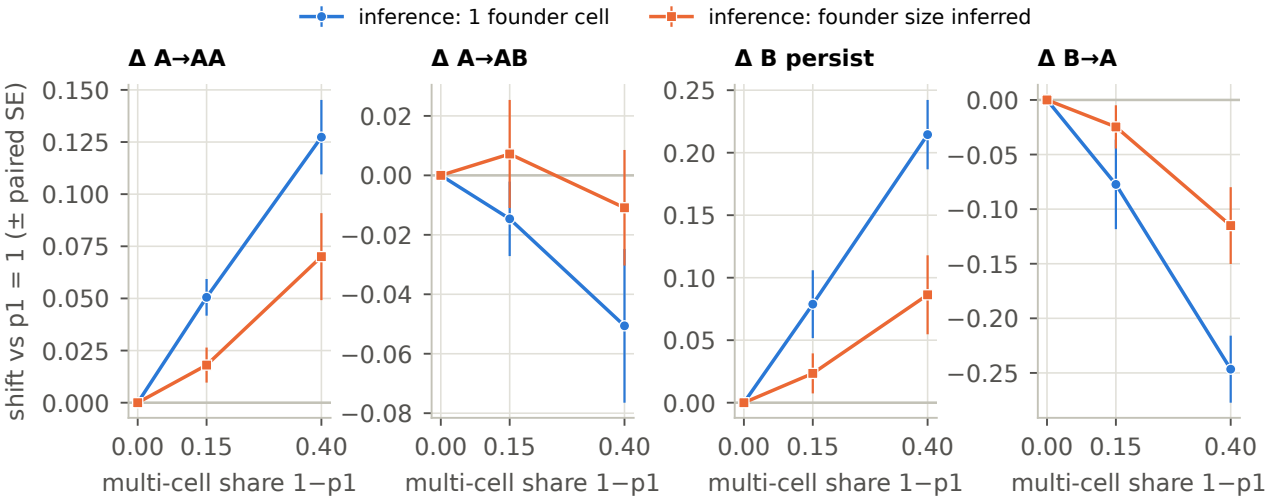


Figure 13: E1a founder number, GW: shift vs the $p_1 = 1$ control of the same arm (± paired SE).

Arm, p_1	A→AA (.55)	B persist (.10)	B→A (.80)	`founder_size.1`	PPC	support (min)
[1], control	0.52 [3/5]	0.13	0.76 [5/5]	fixed	0.004	0.67 (0.44)
[1], 0.85	+0.05 ± .01	+0.08 ± .03	−0.08 [4/5]	fixed	0.042	0.56 (0.21)
[1], 0.6	**+0.13 ± .02** [0/5]	**+0.21 ± .03** [0/5]	**−0.25 ± .03** [0/5]	fixed	**0.117**	**0.16** (0.015)
[1,2,3], control	0.46 [3/5]	0.09	0.83 [5/5]	0.85	0.000	0.55 (0.52)
[1,2,3], 0.85 / 0.6	+0.02 / +0.07 ± .02	+0.02 / +0.09	−0.02 / −0.12 ± .04 [5/5]	0.80 / 0.71	≤ 0.004	0.53 / 0.56

Comment

- **A large, partly detectable error in GW**: 40% multi-cell clones ignored inflate self-renewal (+0.13) and lower B→A (−0.25), all 0/5 covered; the fit is flagged (PPC 0.12, support 0.16). At $p_1 = 0.85$ the shifts are a third of that and the checks borderline.
- **Contrast with BH** (§8): there the same error moved B→A by only −0.07 and was silent — the lifetimes absorb part of the extra growth. A free time scale hides the error.
- **Inferring the founder size helps in GW**: misfit removed (PPC ≤ 0.004), shifts halved and covered, `founder_size.1` tracks p_1 (0.85 / 0.80 / 0.71, contraction ≈ 0.2). Cost on single-cell data: A→AA −0.06 ± 0.01, mean contraction 0.89 → 0.73.
- Consequence: record at seeding whether wells contain multi-cell seeds; if ≳15% of organoids start from several cells, fix the measured distribution or infer it. $p_1 < 0.6$ not tested.

10 E1b — Death rates and the observed-organoid fraction (BH and GW)

Question

How strongly do A death and B death change the fraction of seeded organoids still detected at the endpoints? Is it an informative extra datum?

Setup

Simulator	BH (simulated, rejection loop) and GW (exact $1 - P$(no captured cell))
Law	`reversible_ab`; A death ∈ {0, .03, .1, .2, .3} × B death ∈ {0, .1, .3, .5}, other entries of each row rescaled in proportion
Data vs inference	no inference used (token budget: 256 simulations, 5 epochs; posteriors meaningless)
Design	real-like: BH 144/240 h, GW generations 6/10; 165/101 recovered organoids
Seeds / runs	1 dataset per cell / 40 (20 BH + 20 GW)
Folder	`results/e1b/bh`, `results/e1b/gw`

Result

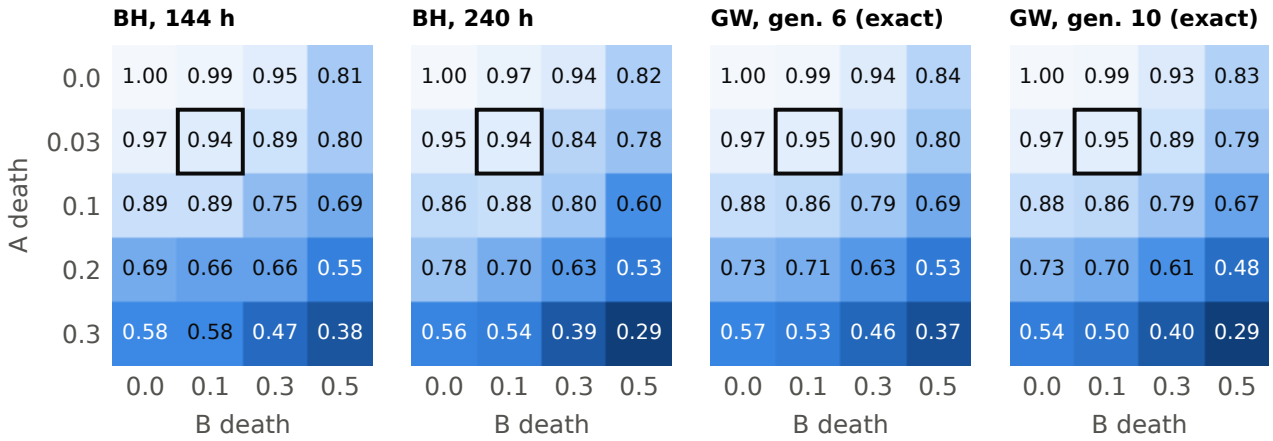


Figure 14: Observed-organoid fraction (detected/seeded) per (A death, B death); box = reference truth. BH: binomial SE ≈ 0.02–0.03 (100–530 seeded clones); GW exact.

(A death, B death)	BH 144 h	BH 240 h	GW gen. 6	GW gen. 10
(0, 0)	1.00	1.00	1.000	1.000
(0.03, 0.1) reference	**0.94**	**0.94**	**0.953**	**0.952**
(0.3, 0)	0.58	0.56	0.565	0.543
(0, 0.5)	0.81	0.82	0.840	0.832
(0.1, 0.3)	0.75	0.80	0.793	0.785
(0.3, 0.5)	0.38	0.29	0.366	0.290

Comment

- A death dominates (0 → 0.3: 1.00 → 0.54–0.58); B death 0 → 0.5 only to ≈ 0.82.
- Losses happen early (day 6 ≈ day 10), except high B death with A death ≥ 0.1.
- Different (A, B) death pairs give the same fraction — confirmed by the exact GW values ((0, 0.5) and (0.1, 0.3): 0.84 vs 0.79; (0.03, 0.5): 0.80): one constraint on a ridge, not two measurements.
- GW exact and BH simulated agree on average to 0.02 (max 0.07, both signs; within ≈ 2 binomial SE of BH) at the same days/generations.
- Consequence: record the number of seeded organoids per well (Q5). A measured fraction well below 0.9 would indicate A death $\gtrsim 0.1$. Not yet tested: using it as a summary (`observation=all_clones`).
- Caveat: death also lowers the growth rate here (rows rescaled).

11 E1c — Death rates fixed to wrong values (BH)

Question

If A and B death are fixed (possibly wrongly) and the rest inferred, do other parameters compensate (flat PPC valley = death not identifiable once fixed)?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	`reversible_ab` with A death, B death fixed (7 free)
Data vs inference	data always $\theta^\star$ (A death 0.03, B death 0.1); inference fixes A death $\in \{0, 0.03, 0.2\}$, B death $\in \{0, 0.1, 0.4\}$
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k (retrained per setting)
Seeds / runs	5 / 45 (3 × 3 grid; the 4 × 4 grid is the GW sweep, §12)
Control	deaths fixed at truth (0.03 / 0.1)
Folder	`results/e1c/bh/`

Result

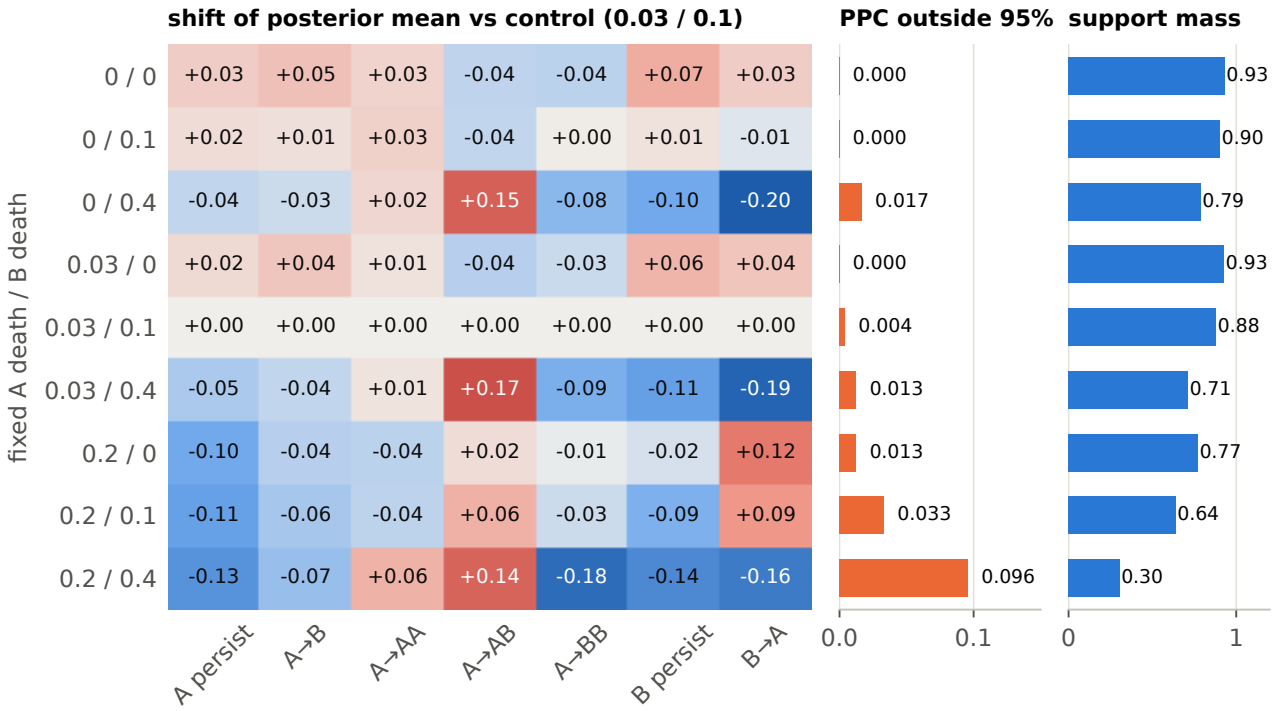


Figure 15: Left: shift of posterior means vs the control (deaths fixed at truth). Right: PPC and posterior support mass (well-specified free-death baseline: support 0.78).

Fixed A / B death	A persist	A→AB	B→A	τ_A	τ_B	PPC	support
0.03 / 0.1 (control)	0.15	0.18	0.67 [5/5]	16.1	18.1	0.004	0.88
0 / 0.4	−0.04	+0.15	−0.20 [0/5]	+0.3	+3.2	0.017	0.79
0.03 / 0.4	−0.05	+0.17	−0.19 [0/5]	−0.3	+2.9	0.013	0.71
0.2 / 0	−0.10	+0.02	+0.12	−2.1	−3.4	0.013	0.77
0.2 / 0.1	−0.11	+0.06	+0.09	−2.7	−2.1	0.033	0.64
0.2 / 0.4	−0.13	+0.14	−0.16 [0/5]	−2.4	+7.9	**0.096**	**0.30**
0 / 0, 0 / 0.1, 0.03 / 0	≤ 0.03	−0.04	≤ 0.04	≤ 0.8	≤ 2.4	0.000	0.90–0.93

Paired SE ≤ 0.03 (probabilities), ≤ 0.35 h (τ_A), ≤ 1.6 h (τ_B).

Comment

- Wrong deaths are absorbed by biologically sensible compensations: B death too high ⇒ less B→A, more A→AB; A death too high ⇒ less quiescence/switching, faster A clock.
- Bias, not information loss: mean contraction *rises* (0.72 → 0.84) while coverage falls (0.76 → 0.38): wrong models are more confident.
- Only the double error is flagged (PPC 0.096, support 0.30; seed 2 support 0.014 ⇒ not an estimate).
- Fixing deaths at the truth gains little vs free deaths (free baseline already constrains them: contraction 0.89 / 0.65).
- Consequence: do not fix death to a guess; keep free or measure (apoptosis marker). C2ST not run; intermediate values (0.1, 0.2) only in the GW sweep (§12).

12 E1c — Death rates fixed to wrong values (GW, 4 × 4 grid)

Question

As §11, on a finer grid and without the BH ridge: is there a flat PPC valley across fixed death values?

Setup

Simulator	Galton–Watson
Law (free)	`reversible_ab` with A, B death fixed (5 free)
Data vs inference	data always $\theta^\star$; inference fixes A death $\in \{0, .03, .1, .2\}$ × B death $\in \{0, .1, .2, .4\}$
Design	real-like generations 6/10 × 165/101
Inference / budget	NPE amortised, 5k (retrained per setting); bounded SBC (relaunched after the first launch hung; 1 SBC case excluded in 1 run)
Seeds / runs	5 / 80
Control	deaths fixed at truth (0.03 / 0.1)
Folder	`results/e1c/gw/`

Result

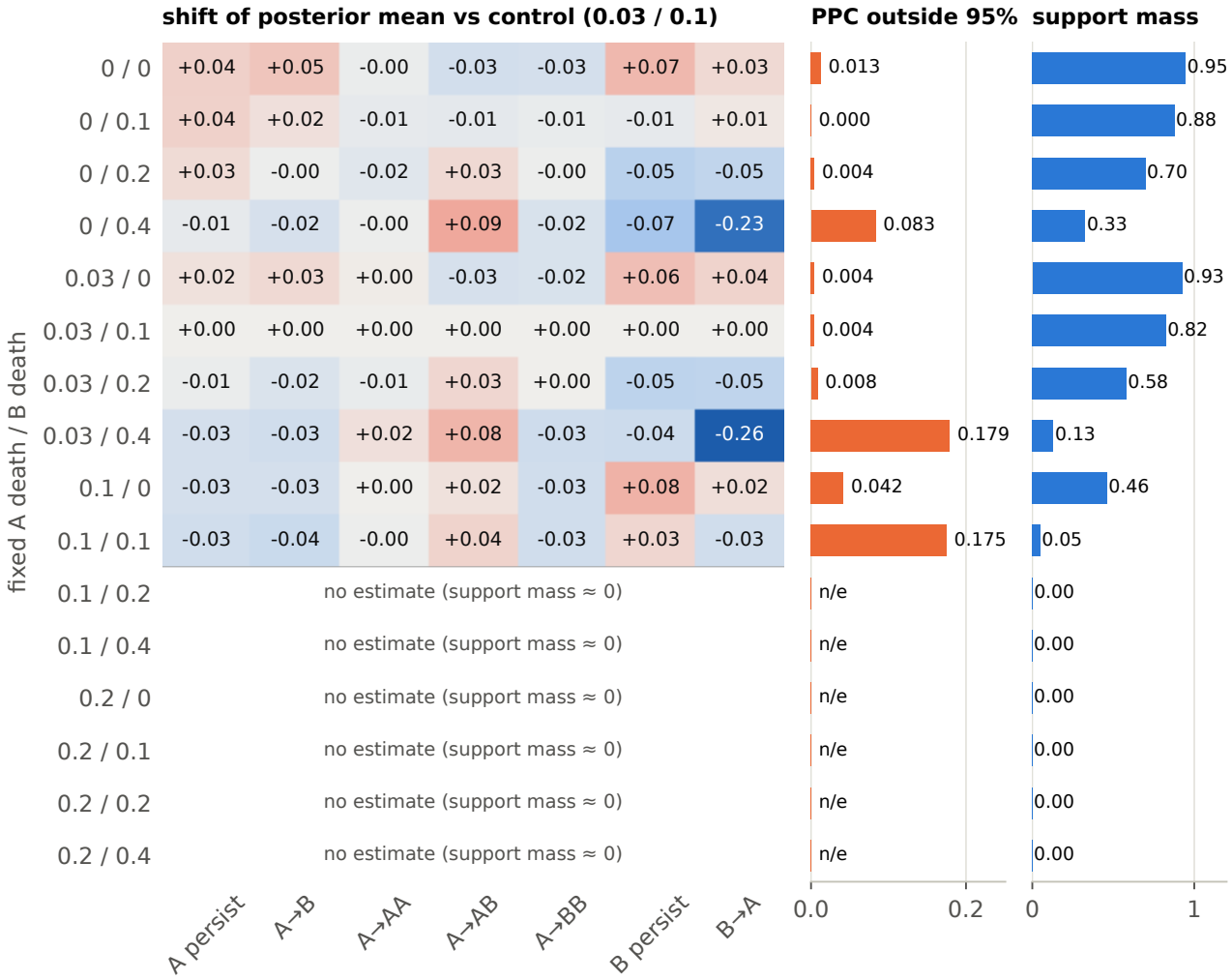


Figure 16: E1c GW. Left: shift of posterior means vs the control (deaths fixed at truth); rows with support mass < 0.02 are not estimates. Right: PPC and posterior support mass.

Fixed A / B death	A persist	A→AB	B persist	B→A	PPC	support
0.03 / 0.1 (control)	0.04	0.10	0.14	0.76 [5/5]	0.004	0.82
0 / 0, 0 / 0.1, 0.03 / 0	$\leq +0.04$	≥ -0.03	$\leq +0.07$	$\leq +0.04$	≤ 0.013	0.88–0.95
0 / 0.2, 0.03 / 0.2	$+0.03$ / -0.01	$+0.03$	-0.05	$-0.05 \pm .02$ [0/5]	≤ 0.008	0.58–0.70
0 / 0.4	-0.01	$+0.09$	-0.07	-0.23 [0/5]	**0.083**	**0.33**
0.03 / 0.4	-0.03 [1/5]	$+0.08$	-0.04	-0.26 [0/5]	**0.179**	**0.13**
0.1 / 0	-0.03 [0/5]	$+0.02$	$+0.08$	$+0.02$	0.042	0.46
0.1 / 0.1	-0.03 [0/5]	$+0.04$	$+0.03$	-0.03	**0.175**	**0.05**
0.1 / 0.2, 0.1 / 0.4	no estimate (support mass ≤ 0.004)				—	**0.00**
0.2 / any	no estimate (17/20 runs without any posterior sample)				—	**0.00**

A→AA never moves ($|\Delta| \leq 0.02$) in an estimable cell. Paired SE ≤ 0.02 for the shifts shown.

Comment

- **The valley is flat downwards, walled upwards for A**: death fixed at 0 fits as well as the truth (shifts ≤ 0.07, PPC ≤ 0.013; small, though up to ≈ 2 posterior sd for GW's narrow posteriors). A death fixed above the truth breaks the fit from 0.1 on (support ≈ 0; at 0.2 no posterior mass in the prior support): GW has one event per generation, so a higher death must be absorbed within A's simplex (A persist and A→B hit 0 first).
- **Contrast with BH** (§11): there A death 0.2 was absorbed silently by a faster A clock (PPC ≤ 0.033, support 0.64–0.77, A persist -0.10, τ_A -2 to -3 h). Whether a wrong fixed death is *detectable* depends on whether the model has a free time scale.
- **B death too high biases B→A in both**: 0.4 ⇒ $-0.23/-0.26$ (GW, flagged: support 0.13–0.33) vs $-0.19/-0.20$ (BH, silent). B death 0.2 is silent in GW but already 0/5 covered (B→A -0.05; with B death fixed, B's row has one free coordinate and a narrow posterior).
- Fixing deaths at truth gains little over free deaths (MAE 0.028 vs 0.033). Consequence unchanged: do not fix death rates to guesses. C2ST not run.

13 E2a / E2b — A↔B mis-typing, known vs ignored (BH)

Question

Clustering mislabels cells at rate e. If the error is modelled (E2a), do we only lose information? If ignored (E2b), how large is the bias, and is it visible?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	`reversible_ab` (9)
Data model	symmetric confusion $e \in \{0, .02, .05, .1, .15, .2\}$, capture 1
Inference model	E2a: `observation=noisy` with the same e (network retrained per e); E2b: perfect typing (one network per seed)
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 60
Control	$e = 0$ (= BH baseline)
Folder	`results/e2/known_bh`, `results/e2/ignored_bh`

Result

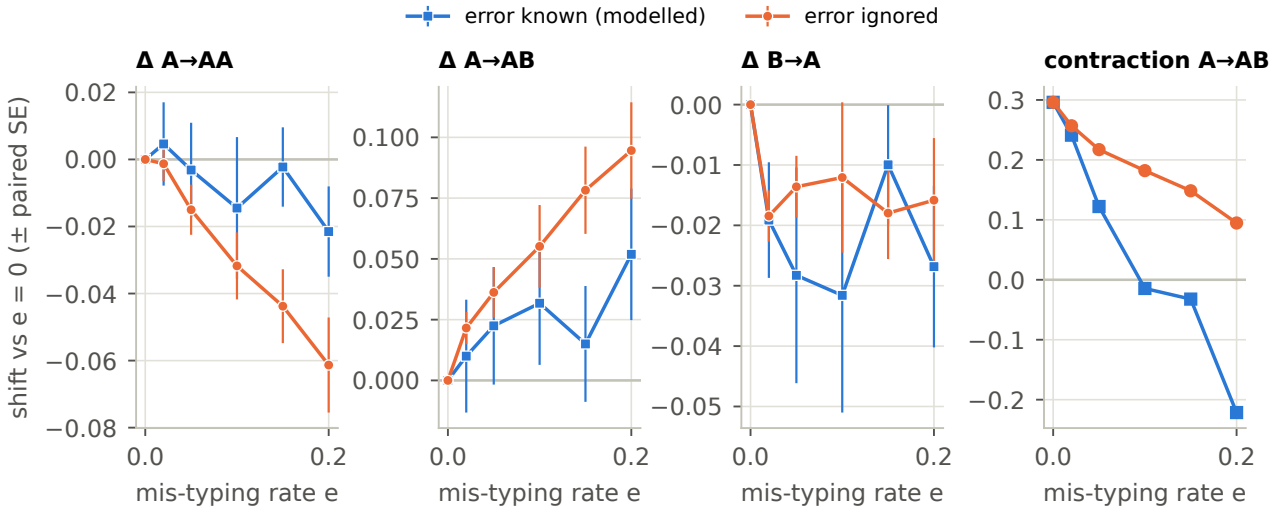


Figure 17: Shift vs $e = 0$ (± paired SE) and contraction of A→AB. Blue: error modelled; orange: error ignored.

e	0.02	0.05	0.1	0.15	0.2	PPC / support
Ignored: A→AB shift	+0.02	+0.04	+0.06	+0.08	+0.09 ± .02	≤ 0.008 / 0.78
Ignored: A→AA shift	−0.00	−0.02	−0.03	−0.04	−0.06± .01	
Ignored: shift / posterior sd (A→AB)	0.2	0.3	0.4	0.6	0.7	
Known: contraction A→AB (0.30 at e=0)	0.24	0.12	−0.01	−0.03	−0.22	≤ 0.008 / n/r
Known: contraction A→AA (0.52)	0.53	0.48	0.43	0.41	0.38	

Comment

- Known error = information loss in the division split; means within 0.05 of control, no trend.
- Ignored error = monotone bias: mislabelling makes clones look mixed, explained by more asymmetric divisions. Coverage of A→AB 3/5 → 0/5 at $e = 0.2$. Invisible to PPC and support mass.
- At the estimated rates (5% ++, 15% −−; Q2) the bias is < 1 sd at this design — tolerable only because posteriors are wide; larger designs (E4) narrow them.
- Consequence: estimate e (2D vs 3D, Q3) and pass it to the model. Asymmetric errors not tested.

14 E2a / E2b — A↔B mis-typing, known vs ignored (GW)

Question

As §13 for Galton–Watson: tolerance to known vs ignored mis-typing when the baseline is unbiased and the posteriors are narrower.

Setup

Simulator	Galton–Watson
Law (free)	`reversible_ab` (7)
Data / inference	symmetric $e \in \{0, .02, .05, .1, .15, .2\}$; modelled (E2a, network retrained per e) or ignored (E2b, one network per seed)
Design	real-like generations 6/10 × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 60
Control	$e = 0$ (= GW baseline)
Folder	`results/e2/known_gw`, `results/e2/ignored_gw`

Result

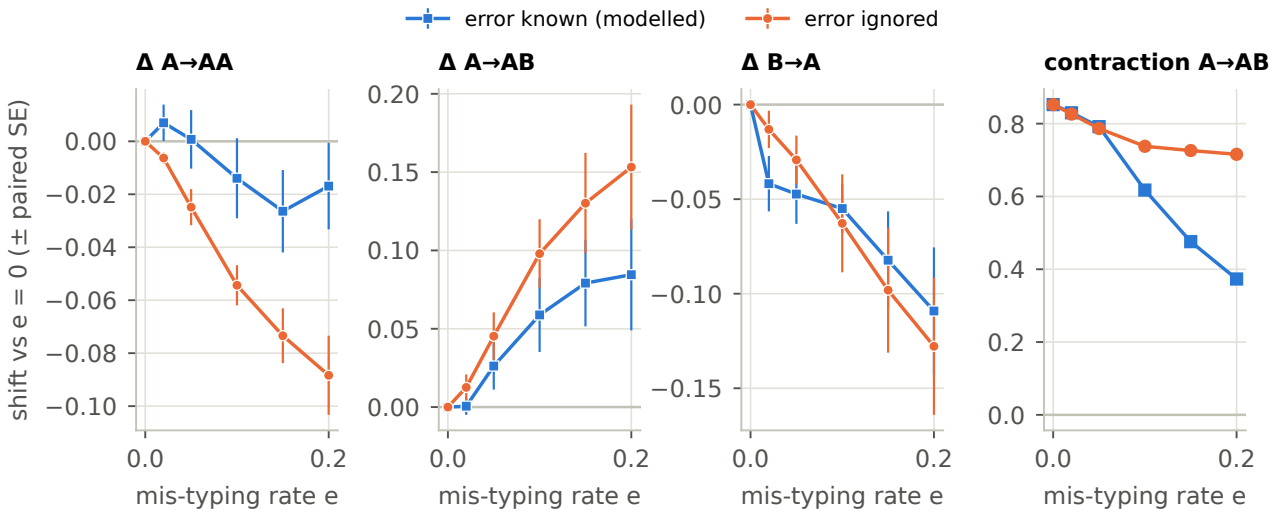


Figure 18: E2 GW: shift vs $e = 0$ (± paired SE) and contraction of A→AB. Blue: error modelled; orange: error ignored.

e	0.02	0.05	0.1	0.15	0.2	BH at 0.2
Ignored: A→AB shift	+0.01	+0.05	+0.10 [0/5]	+0.13	+0.15 ± .04 [0/5]	+0.09
Ignored: A→AA shift	−0.01	−0.02	−0.05 [0/5]	−0.07	−0.09± .01 [0/5]	−0.06
Ignored: B→A shift	−0.01	−0.03	−0.06	−0.10	−0.13± .04 [3/5]	−0.02
Ignored: shift / posterior sd (A→AB)	0.2	0.7	**1.4**	1.8	2.1	0.7
Ignored: PPC / support	.004 / .66	.004 / .64	0 / .59	.012 / .55	.033 / .48	.000 / .78
Known: contraction A→AB (0.85 at e=0)	0.83	0.79	0.62	0.48	0.37	−0.22
Known: B→A shift	−0.04	−0.05	−0.05	−0.08	−0.11± .03 [4/5]	−0.03

Comment

- **Same mechanism as BH**, about 1.5× larger and extending to B's row: mislabelled clones look mixed ⇒ more asymmetric division, less self-renewal, fewer B→A, more B persist (+0.18 at $e = 0.2$).
- **Tolerance is set by the posterior width**: GW posteriors are ≈2.5× narrower (sd of A→AB 0.05 vs 0.12), so the bias exceeds one posterior sd at $e \approx 0.1$ (0/5 covered). Tolerable $e \approx 0.05$ in GW vs ≈ 0.2 in BH.

- **Modelled error = information loss**: A→AB contraction 0.85 → 0.37; means drift towards the prior means (B→A −0.11, A→AB +0.08 at $e = 0.2$) with coverage kept — expected for a wider posterior under a uniform Dirichlet, not a bias.
- **Not detected**: PPC ≤ 0.033; support mass falls 0.67 → 0.48 (min 0.10) — a trend across the sweep, not a single-run warning.
- Consequence: estimate e (Q3) and model it; the BH tolerance will shrink to the GW one once division times are measured. Asymmetric errors not tested.

15 E2c — Hidden third type ignored (GW)

Question

Data contain a third type C (double negative/positive) read as A with prob. q and as B with $1 - q$. If the model ignores C, are the A and B rows still recovered, and does the PPC flag the two-type model?

Setup

Simulator	Galton–Watson
Data model	`three_type`: A = {death .03, persist .02, A→C, A→AA, A→AC, A→CC}, C = {death .05, persist .10, C→B .40, C→BC .20, C→CC .25}, B = {death .10, persist .90} (B never returns to A). C fraction ≈ 5% (A→AA 0.905) or ≈ 15% (A→AA 0.79); `three_split` with $q \in \{0, 0.5, 1\}$
Inference model	two-type `reversible_ab`, perfect typing (7 free)
Truth for scored names	A death .03, A persist .02, A→AA .905 / .79, A→B = A→AB = A→BB = 0, B death .10, B persist .90, B→A 0
Design	real-like generations 6/10 × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 30
Control	*none* in this sweep (no two-type data with the same A row); arms with C modelled: §16
Folder	`results/e2c/ignore_c/`

Result

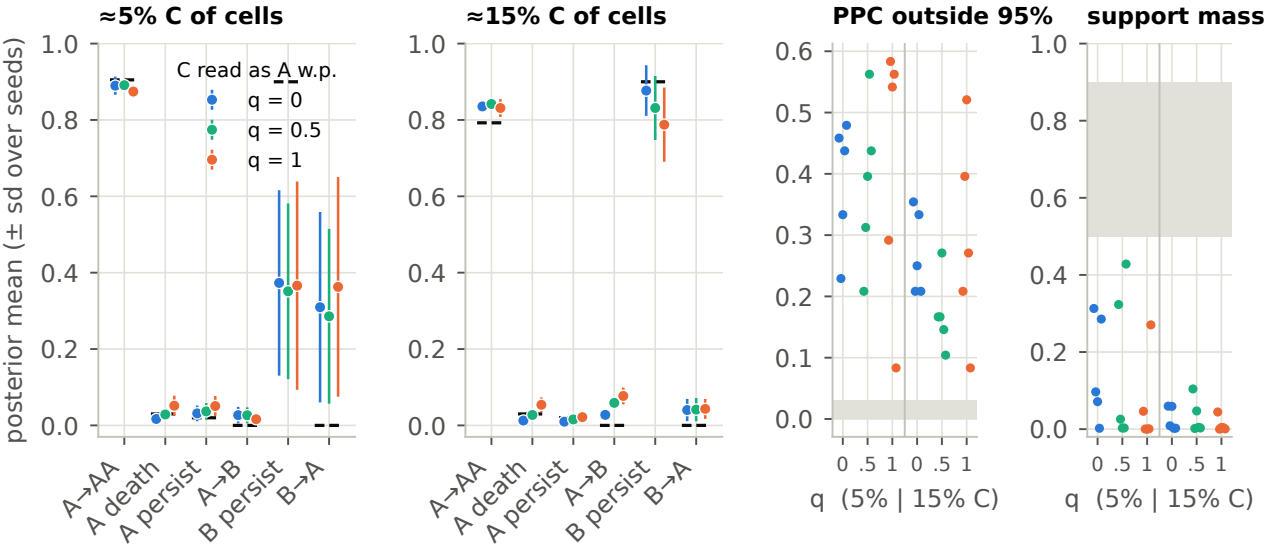


Figure 19: Left: posterior means (± sd over seeds) per q; dashes = truth of the data model. Right: PPC and support mass per run; grey = well-specified ranges.

C share	q	A→AA	A→BB (0)	A→B (0)	B persist (.90)	B→A (0)	PPC	support
≈15%	0	0.84 (.79)	0.10	0.03	0.88	0.04	0.27	0.03
	0.5	0.84	0.05	0.06	0.83	0.04	0.17	0.03
	1	0.83	0.02	0.08	0.79	0.04	0.30	0.01
≈5%	0	0.89 (.905)	0.03	0.03	0.37	0.31	0.39	0.15
	0.5	0.89	0.01	0.03	0.35	0.29	0.38	0.16
	1	0.87	0.01	0.02	0.37	0.36	0.41	0.06

Comment

- **The two-type fit is flagged**: PPC 0.17–0.41 (well specified ≤ 0.03), support mass 0.01–0.16 (well specified 0.5–0.9). Per the reading rules these posteriors are not reliable estimates.
- How C is absorbed: read as B ($q = 0$), A→CC appears as A→BB (0.10 at 15%); read as A ($q = 1$), C→B appears as A→B switching (0.08) and A death rises (0.05). A→AA is overestimated at 15% (+0.05) and slightly under at 5% (−0.02).
- At 5% C, B cells are rare (B only arises via C): B's row stays prior-like (B persist 0.37, contraction 0.25–0.45 for B death).
- **Attribution**: the data truth is far in the prior tail of the inference model (A→AA 0.79–0.905: prior mass above $< 0.05\%$; B persist 0.9: 1%), and PPC is *higher* with less C (5% vs 15%). The arms that model C (§16) fail even more although one of them is the data model: the misfit is driven by the prior tail and cannot be attributed to C.
- One run (5%, $q = 1$, seed 2) drew only 74 posterior samples.

16 E2c — Third type modelled: three-arm comparison (GW)

Question

When C is modelled (freely, or with the trajectory constraint C→B only), are A and B recovered, is C's row identifiable without its own marker, and is the misfit of §15 removed?

Setup

Simulator	Galton–Watson
Data model	as §15 (same 30 datasets: $q \in \{0, .5, 1\}$, ≈5% / 15% C)
Inference arms	ignore C: `reversible_ab` (7 free, §15); C free: `three_type` + `three_split`, q known (27 free) = *the data model*; C trajectory: `three_type_traj` (C never produces A; pending Q1) + `three_split` (23 free)
Design	real-like generations 6/10 × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 30 per arm (90)
Folder	`results/e2c/{ignore_c,model_c_free,model_c_traj}`

Result

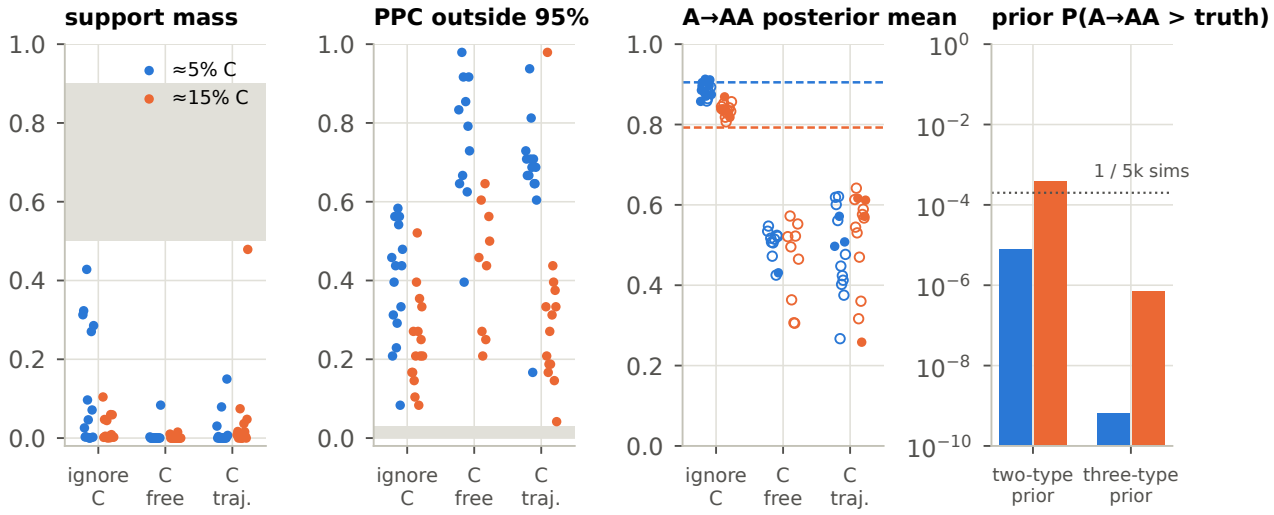


Figure 20: E2c, three inference arms on the same data (one point per run). Grey bands: well-specified ranges. A→AA: open circles = support mass < 0.02 (not estimates); dashed = truth (0.905 at 5% C, 0.79 at 15%). Right: prior probability of A→AA above the truth under each inference model (uniform Dirichlet; dotted = one in 5 000 training simulations).

Arm	C	runs with samples	support: median (max)	PPC (group means)	A→AA (truth)
ignore C	5%	15/15	0.05 (0.43)	0.38–0.41	0.87–0.89 (.905)
	15%	15/15	0.004 (0.10)	0.17–0.30	0.83–0.84 (.79)
C free (well specified)	5%	11/15	0.000 (0.08)	0.63–0.84	*0.48–0.52*
	15%	9/15	0.000 (0.02)	0.37–0.50	*0.33–0.53*
C trajectory	5%	14/15	0.000 (0.15)	0.60–0.75	*0.39–0.60*
	15%	14/15	0.008 (0.48)	0.26–0.38	*0.35–0.59*

Italics: not estimates (support ≈ 0). SBC coverage error 0.024–0.029 in every arm. Prior P(A→AA > truth)

at 5% / 15% C: two-type (6 outcomes) $8 \cdot 10^{-6}$ / $4 \cdot 10^{-4}$; three-type (10 outcomes) $6 \cdot 10^{-10}$ / $8 \cdot 10^{-7}$. P(B persist > 0.9): 0.01 (two-type) vs 10^{-9} (three-type).

Comment

- **Modelling C does not remove the misfit — it makes it worse**: support ≈ 0 in almost every run (10/30 free and 2/30 trajectory runs drew no posterior sample at all), PPC 0.26–0.84. No A/B bias and nothing about C (e.g. C→B 0.01–0.12 vs 0.40) can be read from these arms.
- **The failure is a prior-tail failure, not misspecification**: the free arm *is* the data model. Under a 10-outcome uniform Dirichlet the true A→AA (0.79–0.905) and B persist (0.9) have prior probability 10^{-6}–10^{-10}: no training simulation resembles the data, and amortised NPE extrapolates outside the prior support. SBC is fine because it averages over the prior. The two-type model fits *better* than the true model because its prior has more mass near the truth.
- **Hence the ignore-C flag (PPC 0.17–0.41) cannot be attributed to C**; also within ignore-C the PPC is higher with *less* C. Whether the PPC detects a hidden type is still open. The trajectory constraint helps slightly (fewer empty posteriors; one run with support 0.48).
- Next step (not run): repeat with a truth inside the prior bulk of the three-type model (A→AA $\approx$ 0.4–0.5, C fraction via A→C / A→AC) or an informative prior on A's self-renewal and B's persistence, plus a two-type control with the same A row.

17 E3a — Per-type division time vs one global value (BH)

Question

If B divides more slowly than A, can both lifetimes be inferred? What happens if both are fixed to one global (pooled) value, or to a wrong one (±25%)?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	free arm: `reversible_ab` (9); fixed arm: lifetimes fixed to T (7)
Data model	$\tau_A = 20$ h, $\tau_B \in \{20, 30, 45\}$ h; outcome probabilities $\theta^\star$
Inference model	(i) both lifetimes inferred; (ii) both fixed to pooled $T = (20 + \tau_B)/2$; (iii) pooled ±25%
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 60 (3 τ_B × (1 free + 3 fixed))
Control	free arm at the same τ_B
Folder	`results/e3a/free`, `results/e3a/fixed`

Result

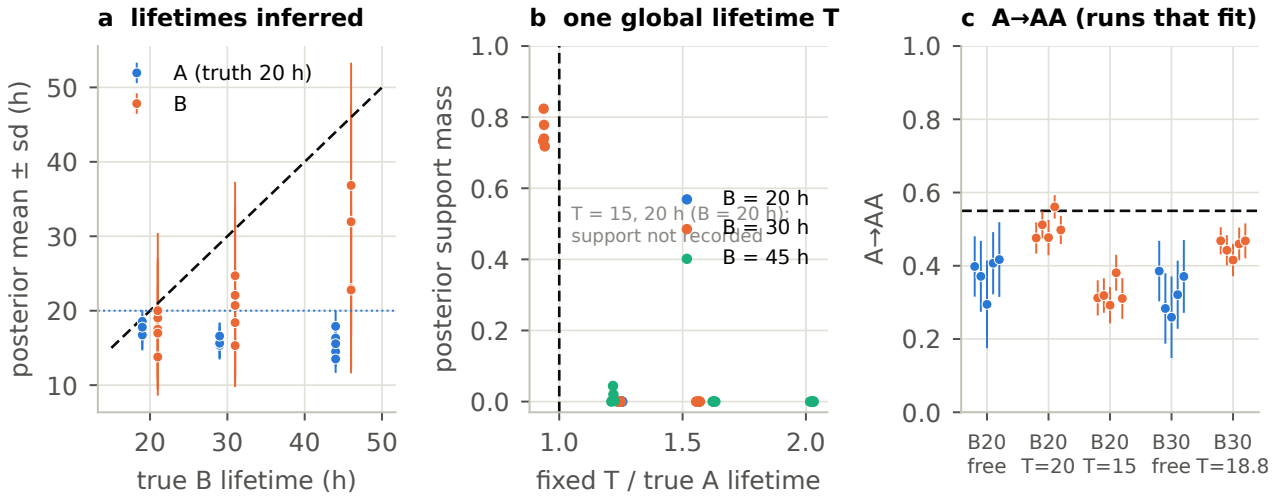


Figure 21: (a) Inferred lifetimes vs true τ_B (per run, ± sd); dotted = true τ_A. (b) Posterior support mass vs fixed T relative to the true τ_A (20 h). (c) A→AA in the settings that fit.

τ_B / T		A persist	A→AA	A→AB	B→A	MAE	PPC	support
20 / free	control	0.11	0.38 [1/5]	0.21	0.68	0.087	0.004	n/r
20 / **20**	T = truth	0.05	**0.50 [4/5]**	0.14	0.75	**0.042**	0.004	n/r
20 / 15	−25%	0.20 (+0.08)	0.32	0.15	0.79 (+0.10)	0.081	0.013	n/r
30 / 18.75	−25%	0.05	0.45 (+0.13)	0.13	0.59 [1/5]	0.072	0.000	0.76
any / ≥ 24.4	$T > \tau_A$	no estimate (0–2 000 samples; boundary values)				—	0.35–0.51	≈ 0

Free arm: τ_A 17.5 / 15.9 / 15.6 h and τ_B 17.5 / 20.2 / 29.3 h for true τ_B 20 / 30 / 45 (contraction of τ_B 0.84 / 0.75 / 0.49).

Comment

- Lifetimes fixed at the truth give the best BH recovery of all (A→AA 0.50, MAE 0.042, contraction 0.92) — direct evidence for the ridge.
- Inferred τ_A is biased short whatever τ_B; τ_B is compressed towards τ_A and weakly informed.
- T longer than A's lifetime ⇒ the model cannot fit (support ≈ 0, PPC 0.35–0.51): *detectable*. Depends on T vs τ_A, not on the pooled mean.
- T a little shorter than τ_A ⇒ fits normally but shifts outcomes by up to 0.10 (A persist absorbs the faster clock): *undetectable*.
- Consequence: measure A's division time per type; never use a pooled mean when B is slower. Caveat: coarse T grid (failure threshold between 20 and 24.4 h).

18 E3b — Desynchronisation: lifetime CV misspecified (BH)

Question

How biased is the fit when the data's lifetime variability (Gamma shape 1–32, CV 1–0.18, or per type) differs from the assumed shape 8 (CV 0.35)? Does the PPC see it?

Setup

Simulator	Bellman–Harris
Law (free)	`reversible_ab` (9)
Data model	shape $\in \{1, 2, 4, 8, 32\}$ and per type {A 4, B 16}, {A 16, B 4}
Inference model	shape 8 always (one network per seed: paired shifts)
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 35
Control	data shape 8 (= BH baseline)
Folder	`results/e3b/`

Result

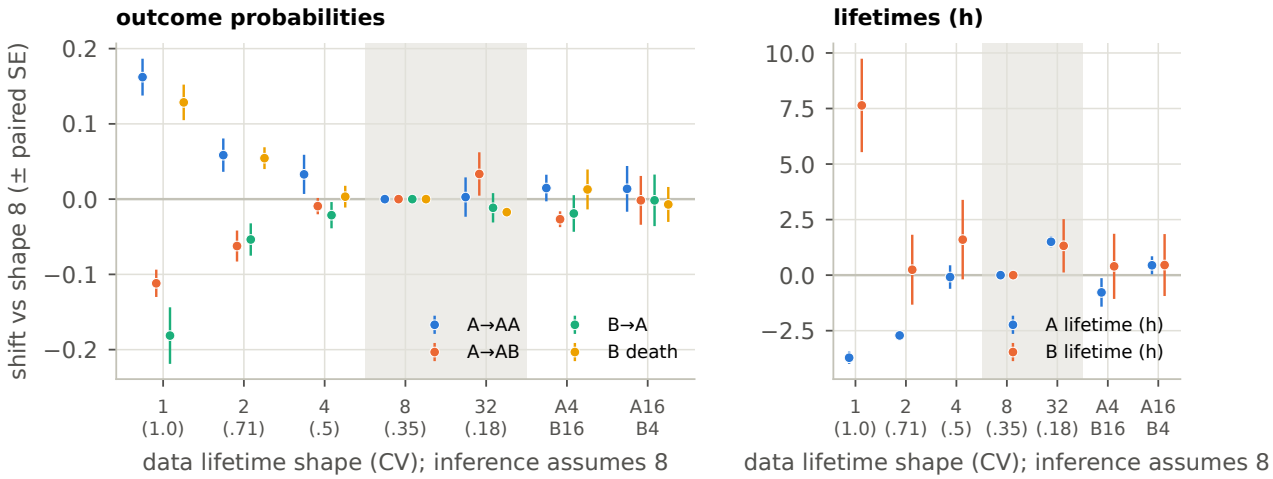


Figure 22: Shift vs shape 8 (± paired SE); grey band = CV 0.18–0.35, around the assumed value.

Data shape (CV)	A→AA	A→AB	B death	B→A	τ_A (h)	τ_B (h)	PPC
8 (0.35) control	0.32	0.23	0.17	0.63	15.9	20.2	0.008
1 (1.0)	+0.16 ± .02	−0.11	+0.13	−0.18± .04	−3.7± 0.3	+7.6 ± 2.1	0.004
2 (0.71)	+0.06	−0.06	+0.05	−0.05	−2.7	+0.2	0.000
4 (0.5)	+0.03	−0.01	+0.00	−0.02	−0.1	+1.6	0.000
32 (0.18)	+0.00	+0.03	−0.02	−0.01	+1.5 ± 0.2	+1.3	0.000
A 4 / B 16; A 16 / B 4	$\leq$ 0.01	$\leq$ 0.03	$\leq$ 0.01	$\leq$ 0.02	−0.8 / +0.4	$\leq$ 0.5	$\leq$ 0.004

Comment

- CV 0.18–0.5 (incl. per-type mixes) is harmless: shifts ≤ 0.035 and ≤ 1.6 h, small vs posterior sd (≈ 0.1, 2 h).
- Near-exponential cycle times are read as a faster clock with a different division split; bias is large and invisible to the PPC. (A→AA moves *towards* the truth only because the control is biased low.)
- Consequence: a rough CV estimate (live imaging) suffices; if CV $\gtrsim 0.7$ change the shape in the model. Shape was never inferred; support mass not recorded.

19 E3c — Quiescence vs division time (BH)

Question

Are quiescence (A persist) and A's lifetime confounded? What happens if quiescence is fixed to 0 while the data have A persist $\in \{0.03, 0.15, 0.30\}$?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	free arm `reversible_ab` (9); fixed arm A persist = 0 (8)
Data model	A persist 0.03 / 0.15 / 0.30, other A entries rescaled (A→AA 0.55 / 0.48 / 0.40)
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 30
Control	free arm at the same data
Folder	`results/e3c/persist_free`, `results/e3c/persist_fixed0`

Result

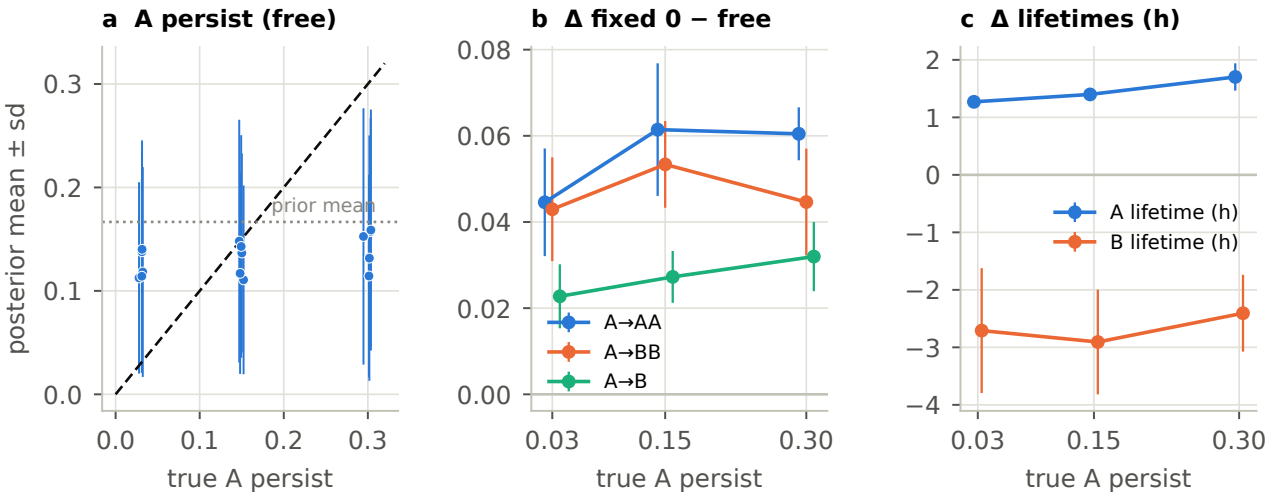


Figure 23: (a) A persist posterior (free) vs truth; dotted = prior mean 1/6. (b, c) Shift fixed-0 minus free (± paired SE).

True A persist	free: A persist (contr.)	free: τ_A	fixed 0: $\Delta\tau_A$	ΔA→AA	$\Delta\tau_B$	PPC
0.03	0.12 (0.51)	15.9	+1.3 ± 0.1	+0.05	−2.7	0.000
0.15	0.13 (0.47)	17.9	+1.4	+0.06	−2.9	0.000
0.30	0.14 (0.35)	19.3	+1.7	+0.06	−2.4	0.004

Posterior correlations (free arm): corr(A persist, τ_A) ≈ −0.6; corr(A→AA, τ_A) ≈ +0.7.

Comment

- Quiescence is not identifiable from endpoint counts: the posterior stays near the prior whatever the truth.
- Quiescence, lifetime and self-renewal lie on one ridge (the growth rate).
- Fixing it to 0 gives a constant small shift (longer lifetimes absorb it), independent of the true value, undetected by the PPC.
- Consequence: fix A persist from independent evidence (Ki67) or report lifetimes and self-renewal conditional on it; measuring A's division time also breaks the confound. Q4 (meaning of "inattivazione") open.

20 Device and batch-size benchmark

Question

Do GPU training and/or a larger batch (512 vs 128) make runs faster without degrading the posterior?

Setup

Configuration	GW, `two_type`, generations 4/8/12 × 32, NPE 20k, seed 0, SBC 200, PPC 200, 4 threads
Hardware	20-core CPU; NVIDIA RTX A1000 (8 GB, also driving the display)
Varied	device (CPU/GPU) × batch (128/512); SNPE 4 rounds on CPU and GPU
Protocol	one job at a time, contention checked every 15 s (all 6 jobs clean); single measurements
Folder	`results/gpu_bench/` (branch `gpu-bench`, `811d8ca`)

Result

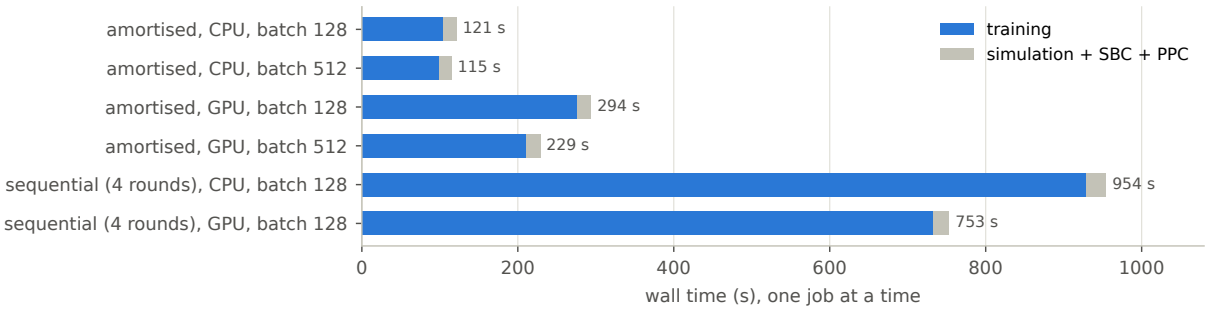


Figure 24: Wall time per job (blue = network training).

Job	wall (s)	s/epoch	epochs	MAE	SBC error
amortised, CPU, 128	121	1.47	71	0.140	0.032
amortised, CPU, 512	115	0.65	151	0.140	0.028
amortised, GPU, 128	294	4.07	68	0.137	0.033
amortised, GPU, 512	229	1.27	166	0.138	0.031
SNPE, CPU, 128	954	4.3	215	0.135	—
SNPE, GPU, 128	753	3.2	226	0.129	—

Comment

- Amortised NPE: GPU 2–2.4× slower (small flows, launch overhead). Batch 512: epochs 2.3× faster but 2× more epochs: −5%.
- SNPE: GPU 21% faster. Quality identical across devices.
- Keep CPU + batch 128; GPU only for SNPE or as extra capacity. These are the only valid timings in this report.

Part II

Planned experiments

All sweeps of E0–E4 are finished (Part I). The experiments below were never run; no results exist for them.

21 X2 — Seed spread: dataset vs training

Question

How much of the seed-to-seed spread comes from the dataset and how much from network training?

Setup

Configuration	final configuration from E1–E4 (to be chosen)
Arms	training spread: 1 dataset × 5 training seeds; dataset spread: 5 `data_seed` × 1 training seed
Runs	5 per arm (planned, not started)
Folder	`results/x2/training`, `results/x2/data`

Result pending — planned

Will show the posterior mean and sd per parameter in both arms. If the training spread is comparable to the dataset spread, single-run results need ensembles or larger budgets.

22 Other planned work (X1, X3, X4, R, P0, follow-ups)

Question

- **X1 methods**: NPE vs NLE vs NRE (and SNPE) at the baseline and the worst realistic error.
- **X3 model**: can a GW fit describe BH data? (needs a GW↔hours mapping; code change).
- **X4 combined errors**: realistic combination (organoid death + mis-typing + C + lifetime CV), real-like design, `budget=full`: the error budget to quote with the real-data fit.
- **R real data**: CRC0322, day 6 (165 organoids) and day 10 (101); choose 2D counts vs 3D estimates (Q3); final model, SBC, fit, PPC.
- **P0 BH timing**: seconds per BH run at `dev`/`smoke` budgets on an idle machine.
- **Follow-ups of finished sweeps**: E2c with a truth inside the prior bulk of the three-type model (or an informative prior) and a two-type control (§16); asymmetric mis-typing (E2); E4 GW with `law=two_type` at 256–512 clones; observed fraction as a summary (`observation=all_clones`, E1b).

Setup

X1	`+experiment=compare_methods`; NLE/NRE with 50 SBC cases; 3 seeds
X4	real-like design, 5 seeds, `budget=full`
R	`submodules/GW/data/df_6D_withmapping.csv`, `df_10D_withmapping_new.csv`
Blocking	Q1 (C trajectory), Q2 (C proportions), Q3 (2D vs 3D mislabelling), Q4 (meaning of inactivation), Q5 (seeded organoids known?)

Result pending — planned, not started

X1: error/contraction/SBC per method. X3: bias of a GW fit to BH data. X4: total bias vs posterior sd per parameter. R: posterior on CRC0322 with the error budget of X4.

A Reproducibility

Code versions. `manifest.json` in each run records `git_commit` (HEAD at run time; since `6b45d23` also the commit of the loaded code). Nearly all runs have `git_dirty = true`. Several sweeps span several HEAD values because commits were made while they ran; the loaded code is that of the launch commit (verified for GW E1a: identical networks across settings per seed).

Experiment	Launch commit	Command (preset; `hydra.sweep.dir=results/...`)
E0	`194e31f`, `b3cbae0`	`+experiment=b_row_identifiability law=two_type|reversible_ab; inference.method.rounds=4; +experiment=b_row_flat_prior`
Design comparison	`b3cbae0`	`+experiment=design_comparison` (laws two_type, reversible_ab)
Device benchmark	`811d8ca` (gpu-bench)	`scripts/run_device_benchmark.sh`
E4 BH	`43e9e44`	`+experiment=e4_design simulator=bellman_harris design.horizons=[240],[144,240],[96,144,240],[144,240,336]`
BH ridge	`19a10ea`	`+experiment=bh_ridge` (+ lifetimes fixed / A concentration)
E1a type	BH `27b1d61`; GW `19a10ea/d46b4c2`	`+experiment=e1a_founder_type simulator=...` (fixed; inferred with `'~law.fixed.founder'`)
E1a number BH	`27b1d61`	`+experiment=e1a_founder_number simulator=bellman_harris`
E1b	BH `43e9e44`; GW `d46b4c2`	`+experiment=e1b_death_detection simulator=...`
E1c BH	`43e9e44/c919397`	`+experiment=e1c_fixed_death simulator=bellman_harris law.fixed.A.death=0,0.03,0.2 law.fixed.B.death=0,0.1,0.4`
E2a/E2b BH	`27b1d61`–`c919397`	`+experiment=e2a_typing_known`, `+experiment=e2b_typing_ignored`
E4 GW	`19a10ea`	`+experiment=e4_design`
E1a number GW	`b0789d3`	`+experiment=e1a_founder_number simulator=galton_watson`
E1c GW	`d46b4c2`	`+experiment=e1c_fixed_death` (4 × 4 grid; relaunched with bounded SBC)
E2a/E2b GW	`12c0827`	`+experiment=e2a_typing_known|e2b_typing_ignored simulator=galton_watson`
E2c ignore C	`19a10ea`	`+experiment=e2c_third_type`
E2c C modelled	free `d46b4c2`; traj. `12c0827`	`+experiment=e2c_third_type law=three_type|three_type_traj observation=three_split`
E3a / E3b / E3c	`27b1d61`–`c919397`	`e3a_lifetime_free`, `e3a_lifetime_fixed`, `e3b_desync`, `e3c_persist`

Every command has the form

```
uv run bio-sbi -m <overrides> torch_threads=4 workers=4 hydra.sweep.dir=<folder>
```

The exact command, grid and fixed settings are in each folder's `info.md`; each run's `.hydra/config.yaml` reproduces it. Packages: `bio-sbi` 0.2.0, `sbi` 0.27.0, `torch` 2.14.1+cu126, `numpy` 2.5.4, `numba` 0.68.0, `hydra-core` 1.3.8, Python 3.12.13.

Rebuilding this report (from the repository root):

```
uv run python report/make_figures.py      # figures/*.pdf, tables/*.csv
cd report && /home/matteo/miniconda3/envs/latex/bin/tectonic report.tex
```

`make_figures.py` only reads `results/`; incomplete sweeps would be skipped and shown as pending boxes. Numbers in tables come from the `info.md` analyses in `results/` (overview: `results/overview.md`) or from `scripts/summarize.py` functions (`mae_probabilities`, per-parameter rows); plotted numbers are in `report/tables/`.

B Glossary

A, B (C)	LGR5$^+$ stem-like, KRT20$^+$ differentiated (hidden third type)
Amortised NPE	one network trained on prior simulations, usable for any dataset
BH / GW	Bellman–Harris (continuous time, lifetimes) / Galton–Watson (generations)
C2ST	classifier two-sample test; accuracy 0.5 = indistinguishable
Clone / organoid	all descendants of one seeding event; recovered if ≥ 1 cell detected
Contraction	$1 - \mathrm{Var}_{\mathrm{post}}/\mathrm{Var}_{\mathrm{prior}}$
Control (paired)	well-specified run of the same sweep with the same seeds and datasets
Coverage (90%)	seeds whose 90% credible interval contains the truth
Design	endpoints (horizons) $\times$ organoids per endpoint
Endpoint / horizon	destructive measurement time (generations for GW, hours for BH)
Founder	seeded cell; `founder` = type mix, `founder_size` = cells per clone
Lifetime τ_A, τ_B	mean time from birth to the cell's event (BH), Gamma with shape 8
MAE	mean absolute error of posterior means over the outcome probabilities
Observed fraction	detected / seeded organoids
Paired SE	SE over seeds of the difference misspecified − control
PPC	posterior predictive check; fraction of summaries outside the 95% predictive interval
Ridge	direction in parameter space along which the data are (nearly) constant
SBC	simulation-based calibration over prior draws
SNPE	sequential NPE (rounds focus on the observed data)
Support mass	share of the NPE posterior inside the prior support
$\theta^\star$	reference truth (Table 1)
Well specified	data model = inference model